AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre SSC-II Mathematics Annual Examinations 2023

Introduction

This document has been prepared for the teachers and candidates of Secondary School Certificate (SSC) Part II (Class X) Mathematics. It contains comments on candidates' responses to the 2023 SSC-II Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

Candidates performed really well in some concepts, such as, Partial Fractions, Linear Equations, Coordinate Geometry, Circles and Pythagoras Theorem. However, candidates who did not score well mostly failed to understand the demands of the questions, often misinterpreting the command words and the stimuli.

Mentioned below are few concepts that teachers need to focus so that the candidates may perform better.

- Basic Statistics
- Linear Inequalities
- Graph Plotting of the Linear Equation
- Tangents to a Circle, Chords and Arcs

Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.

DETAILED COMMENTS
Constructed Response Questions (CRQs)

<table>
<tr><th colspan="2">Question No. 1</th></tr>
<tr><td>Question Text</td><td>At a camping point, 126 tourists visited in a day. Their age groups were recorded in the given table.
<table>
<tr><th>Age Groups</th><th>Number of Tourists (f)</th><th></th><th></th></tr>
<tr><td>16 - 20</td><td>10</td><td></td><td></td></tr>
<tr><td>21 - 25</td><td>25</td><td></td><td></td></tr>
<tr><td>26 - 30</td><td>30</td><td></td><td></td></tr>
<tr><td>31 - 35</td><td>15</td><td></td><td></td></tr>
<tr><td>36 - 40</td><td>20</td><td></td><td></td></tr>
<tr><td>41 - 45</td><td>14</td><td></td><td></td></tr>
<tr><td>46 - 50</td><td>12</td><td></td><td></td></tr>
<tr><td>Total</td><td></td><td></td><td></td></tr>
</table>
Label the given columns and complete the given table. Also, find the arithmetic mean for the given data.</td></tr>
<tr><td>SLO No.</td><td>14.3.2</td></tr>
<tr><td>SLO Text</td><td>Calculate (and interpret) the arithmetic mean by direct method (only), median and mode for grouped data.</td></tr>
<tr><td>Max Marks</td><td>3</td></tr>
<tr><td>Cognitive Level</td><td>A*</td></tr>
<tr><td>Checking Hints</td><td>1 mark for labelling and filling each column (2 marks total).
1 mark for calculating mean by using formula.</td></tr>
<tr><td>Overall Performance</td><td>Few of the candidates filled out the given table with accuracy, demonstrating a sound understanding of the fundamental elements essential for computing the arithmetic mean. However, it is worth mentioning that a many of the candidates encountered difficulties in accurately identifying and performing calculations involving specific variables, namely class marks and the product of frequency and class marks.</td></tr>
<tr><td>Description of Better Responses</td><td>The better responses exhibited a comprehensive understanding of the arithmetic mean of grouped data, along with a thorough grasp of the essential steps required to fulfill the given question's requirements. The candidates proficiently and accurately completed the columns for the mid-point (x) and the product of frequency and mid-point (fx). Additionally, they correctly applied the appropriate formula for calculating the arithmetic mean and demonstrated precise execution of all calculations.</td></tr>
</table>

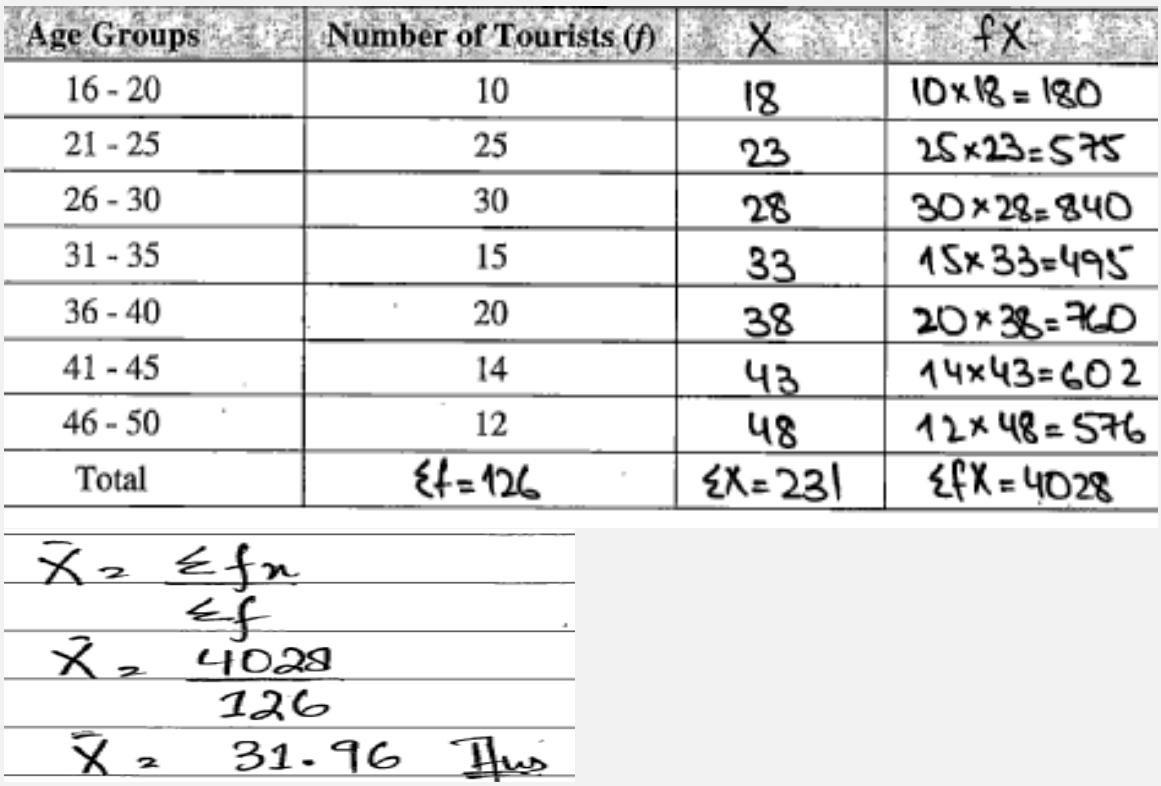
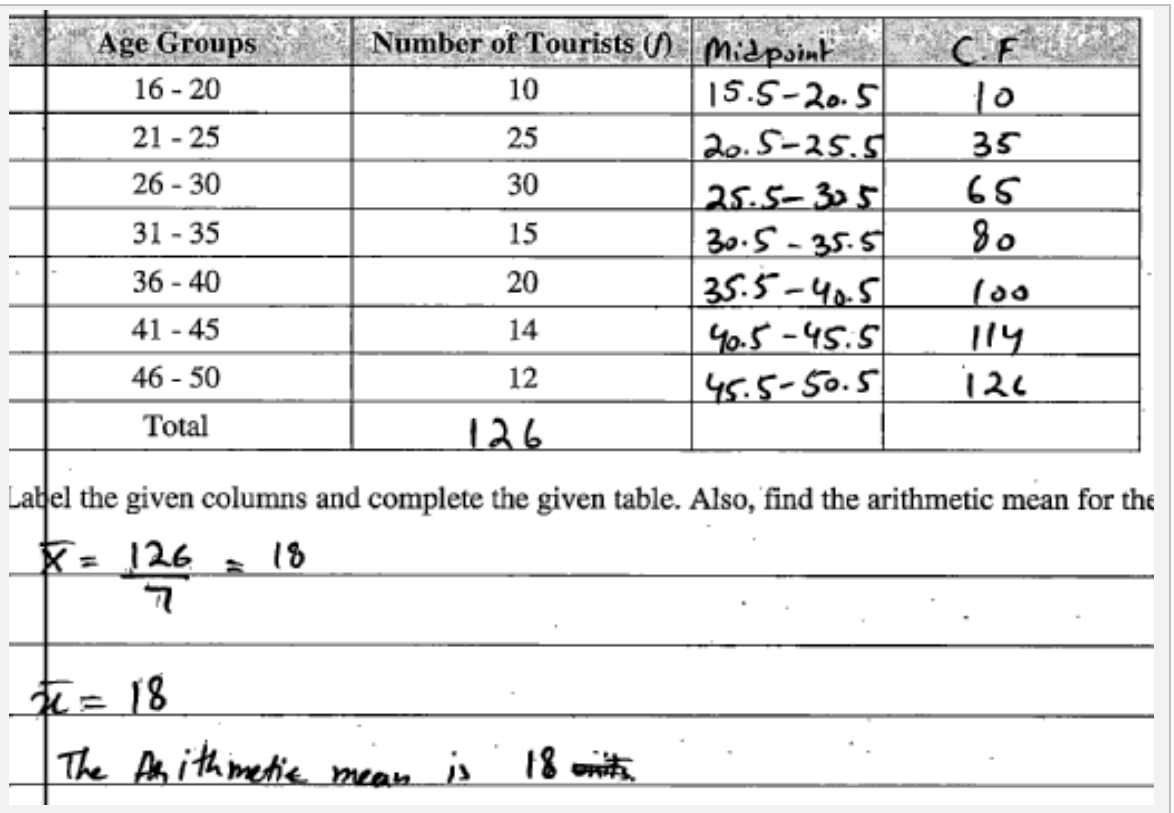
Image of Better Response	
Description of Weaker Responses	Weaker responses from candidates attempting the question indicated the following reasons for their challenges: Misunderstanding of Required Information: Many candidates displayed confusion regarding the information that needed to be filled in the columns of the table. Some struggled to grasp the essential data points required for arithmetic mean calculations, hindering their ability to proceed accurately. Irrelevant Focus on Class Boundaries and Cumulative Frequency: Several candidates deviated from the correct approach by attempting to locate class boundaries or cumulative frequencies, which are not relevant when calculating the arithmetic mean for grouped data. This misdirection led to errors and inaccuracies in their responses. Errors in Applying the Arithmetic Mean Formula: A notable subset of candidates encountered difficulties in correctly applying the arithmetic mean formula. They made errors during the calculation process, which adversely affected the accuracy of their final results.

Image of Weaker Response	

Age Groups	Number of Tourists (f)	Midpoint	C.F
16 - 20	10	15.5-20.5	10
21 - 25	25	20.5-25.5	35
26 - 30	30	25.5-30.5	65
31 - 35	15	30.5-35.5	80
36 - 40	20	35.5-40.5	100
41 - 45	14	40.5-45.5	114
46 - 50	12	45.5-50.5	126
Total	126		

Label the given columns and complete the given table. Also, find the arithmetic mean for the

$\bar{x} = \frac{126}{7} = 18$

$\bar{x} = 18$

The Arithmetic mean is 18 ~~units~~

Suggestions for Improvement (Highlighted part)

How to Approach SLO	Pedagogy** Used for that SLO	Assessment Strategies
• Understand the expectations of the command words • Look at the cognitive level • Identify the content that is required to answer that question (both in terms of understanding of concepts and any skills that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Story Board • Cause and Effect • Fish and Bone • Concept mapping • Audio Visual resources • Think, pair and share • Questioning Technique (Socratic approach) • Practical Demonstration ** For description of each pedagogy, refer to Annexure A	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:
Teachers may implement these teaching methodologies to develop a stronger foundation in arithmetic mean. Use real-world examples and practical scenarios to illustrate the need for calculating arithmetic mean and how it applies to grouped data.

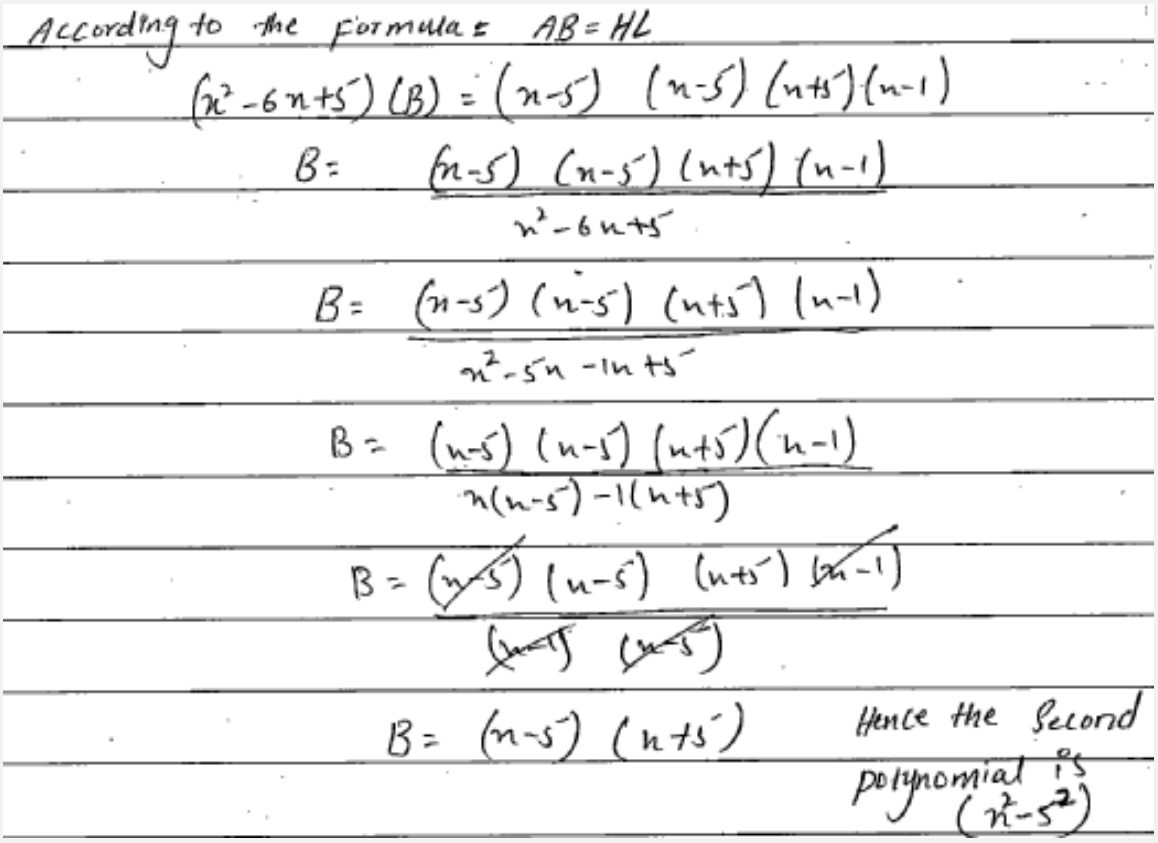
Divide the calculation process into smaller, manageable steps, ensuring students understand each step before moving on.
Emphasise the importance of identifying and collecting the necessary data points, such as class marks and frequencies.
Utilise visual aids, charts, and graphs to help students visualise grouped data and understand its structure.
Conduct interactive activities that involve class participation, such as group discussions, where students can collectively solve arithmetic mean problems for grouped data.
Highlight common errors made during arithmetic mean calculations, such as incorrect formula usage or misinterpretation of data.
Showcase real-life applications of arithmetic mean in fields like economics, statistics, and research to demonstrate its relevance and importance.
Encourage students to explore and discuss how arithmetic mean is used in various professions and industries.

*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

Question No. 2a Candidates were given the choice to attempt any ONE out of the two questions: 2a and 2b.	
Question Text	The HCF and LCM of two polynomials is $x-5$ and $(x-5)(x+5)(x-1)$ respectively. If one of the polynomials is x^2-6x+5, then find the other polynomial.
SLO No.	15.1.3
SLO Text	Apply H.C.F., L.C.M. and their relationship in solving problems.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for factorisation of p(x) 1 mark for substituting the HCF, LCM and p(x) in the relation 1 mark for simplification 1 mark for writing the polynomial in product form i.e. x^2-25.
Overall Performance	This question was related to H.C.F., L.C.M. and their relationship. Most of the candidates successfully scored full marks.
Description of Better Responses	This question was related to HCF and LCM and applying the relation $\text{HCF}\times\text{LCM} = p(x)\times q(x)$. Better responses fulfilled all the requirements of question and were awarded full marks. Candidates exhibited the understanding of the application of LCM by HCF. They substituted the correct expression for HCF, LCM and one of the polynomial to find the 2nd polynomial. They also found the correct factor of x^2-6x+5 and did the simplification of appropriately.

Images of Better Responses

Image (i)



According to the formula = $AB = HL$

$(x^2-6x+5)(B) = (x-5)(x-5)(x+5)(x-1)$

$B = \frac{(x-5)(x-5)(x+5)(x-1)}{x^2-6x+5}$

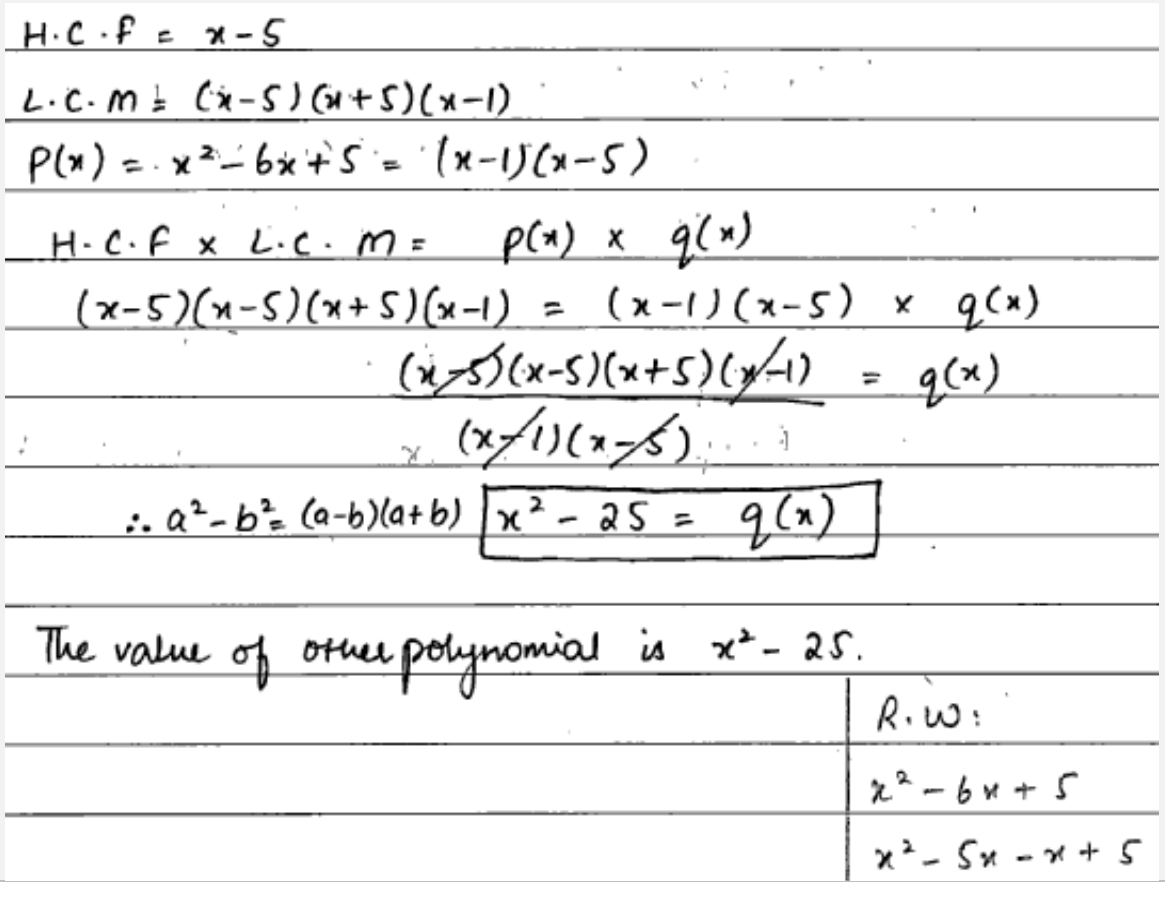
$B = \frac{(x-5)(x-5)(x+5)(x-1)}{x^2-5x-1x+5}$

$B = \frac{(x-5)(x-5)(x+5)(x-1)}{x(x-5)-1(x+5)}$

$B = \frac{(x-5)(x-5)(x+5)(x-1)}{(x-1)(x-5)}$

$B = (x-5)(x+5)$ Hence the second polynomial is (x^2-5^2)

Image (ii)



H.C.F = $x-5$

L.C.M = $(x-5)(x+5)(x-1)$

$P(x) = x^2-6x+5 = (x-1)(x-5)$

H.C.F × L.C.M = $P(x) \times q(x)$

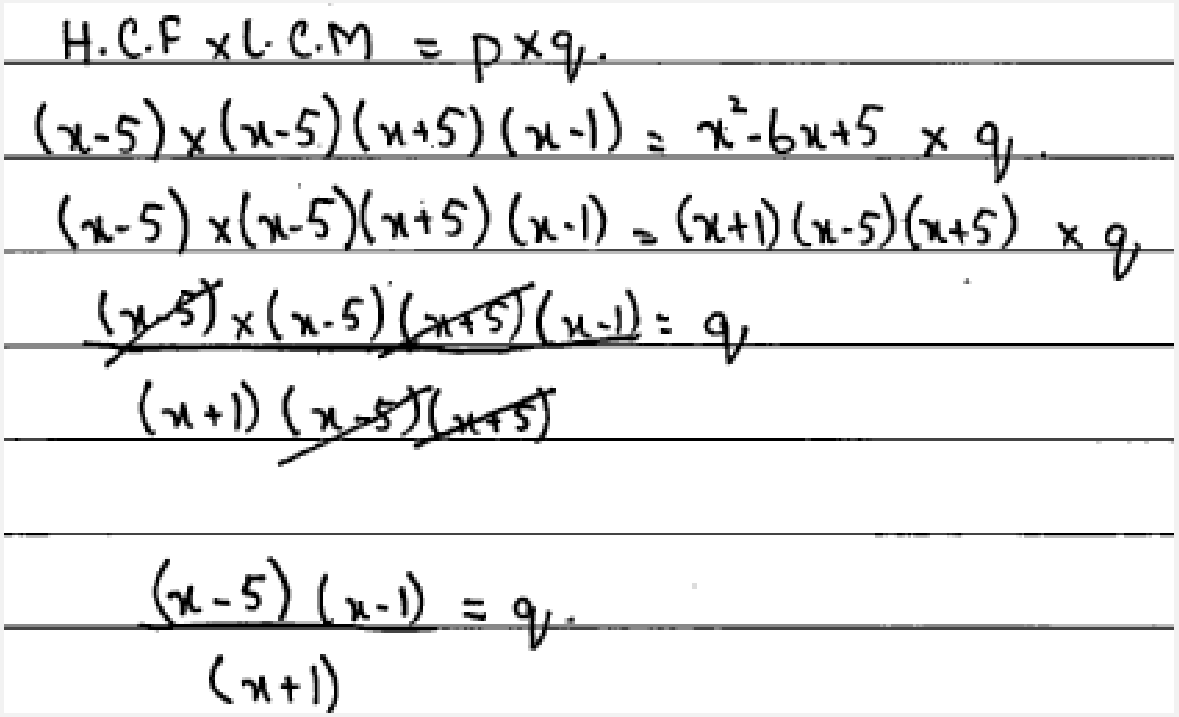
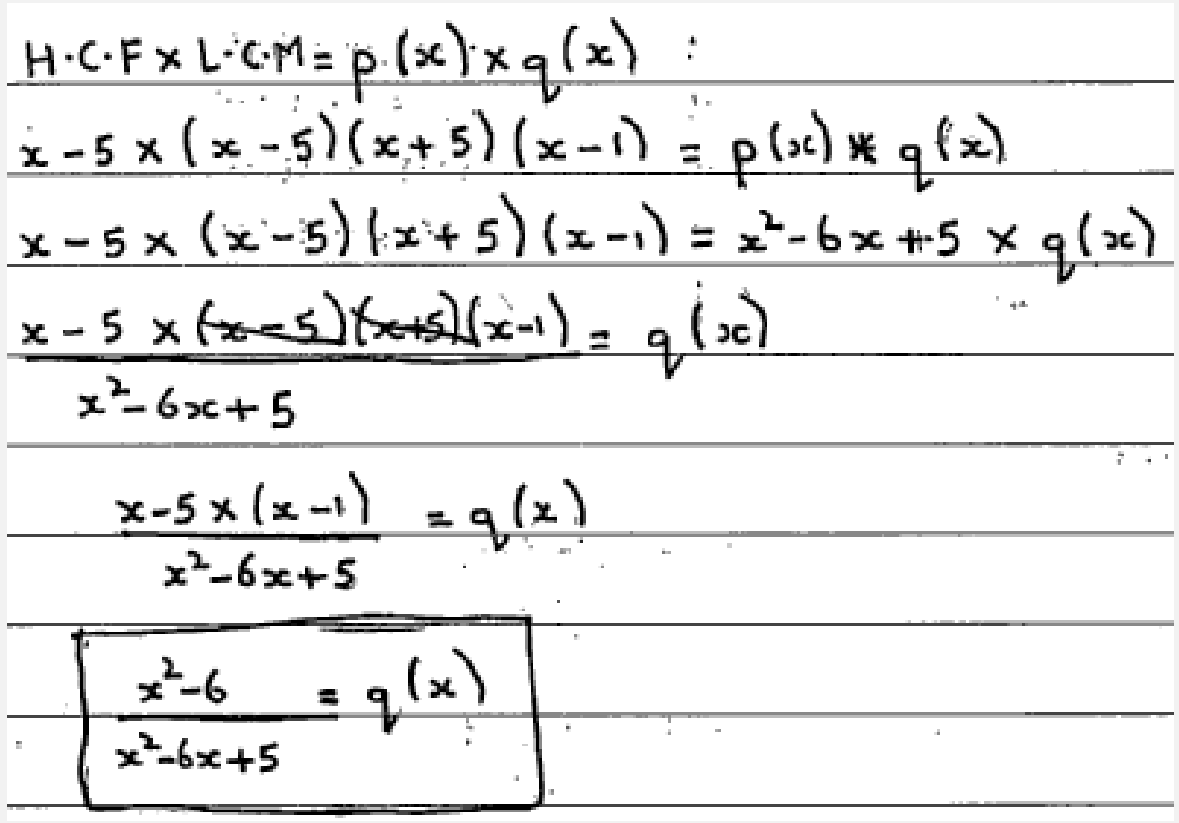
$(x-5)(x-5)(x+5)(x-1) = (x-1)(x-5) \times q(x)$

$\frac{(x-5)(x-5)(x+5)(x-1)}{(x-1)(x-5)} = q(x)$

$\therefore a^2-b^2 = (a-b)(a+b)$ $\boxed{x^2-25 = q(x)}$

The value of other polynomial is x^2-25.

R.W:
x^2-6x+5
$x^2-5x-x+5$

Description of Weaker Responses	Weaker responses indicated the misconception of factorisation of quadratic equation and simplification of algebraic factors. The candidates were unable to demonstrate a clear understanding of using LCM alongside HCF. They did not substitute the correct expressions for HCF, LCM, and one of the polynomials. As a result, they were unable to find the second polynomial accurately.
Images of Weaker Responses	**Image (i)**  H.C.F x L.C.M = p x q. $(x-5)\times(x-5)(x+5)(x-1) = x^2-6x+5 \times q$ $(x-5)\times(x-5)(x+5)(x-1) = (x+1)(x-5)(x+5) \times q$ $\frac{(x-5)\times(x-5)(x+5)(x-1)}{(x+1)(x-5)(x+5)} = q$ $\frac{(x-5)(x-1)}{(x+1)} = q$ **Image (ii)**  H·C·F x L·C·M = p(x) x q(x) $x-5 \times (x-5)(x+5)(x-1) = p(x) \times q(x)$ $x-5 \times (x-5)(x+5)(x-1) = x^2-6x+5 \times q(x)$ $\frac{x-5\times(x-5)(x+5)(x-1)}{x^2-6x+5} = q(x)$ $\frac{x-5\times(x-1)}{x^2-6x+5} = q(x)$ $\frac{x^2-6}{x^2-6x+5} = q(x)$

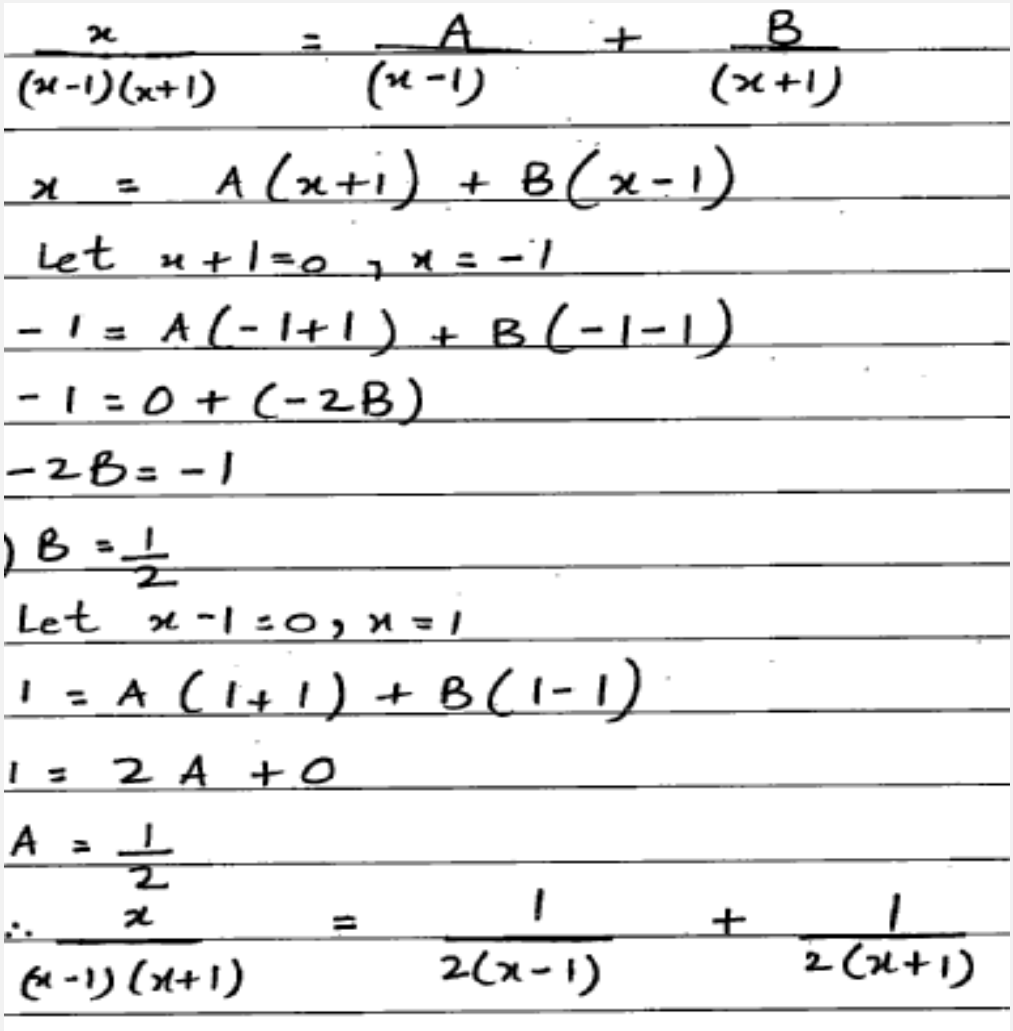
Suggestions for Improvement (Highlighted part)

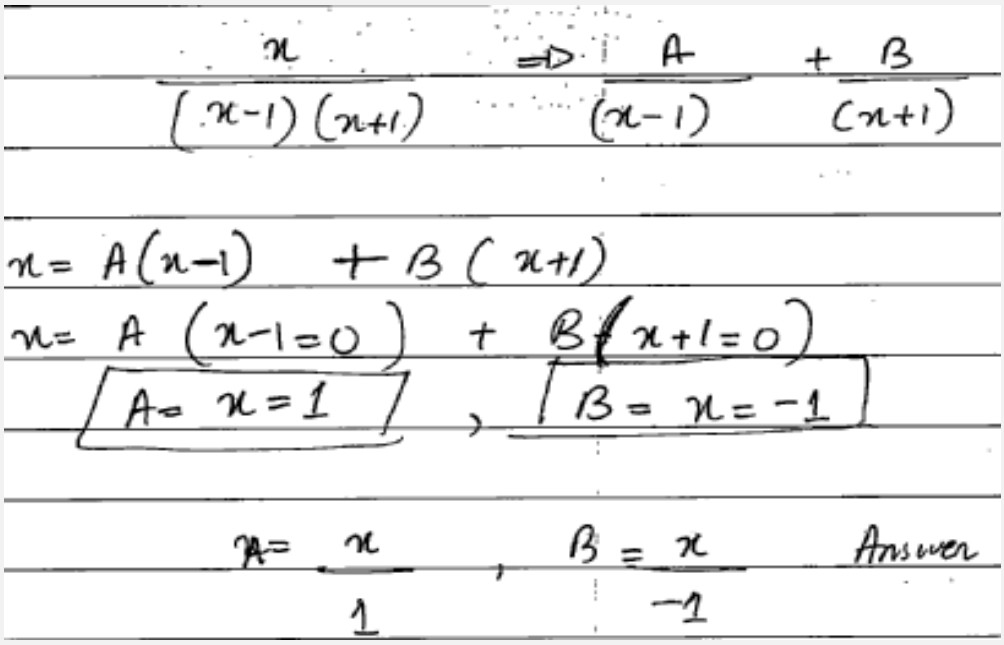
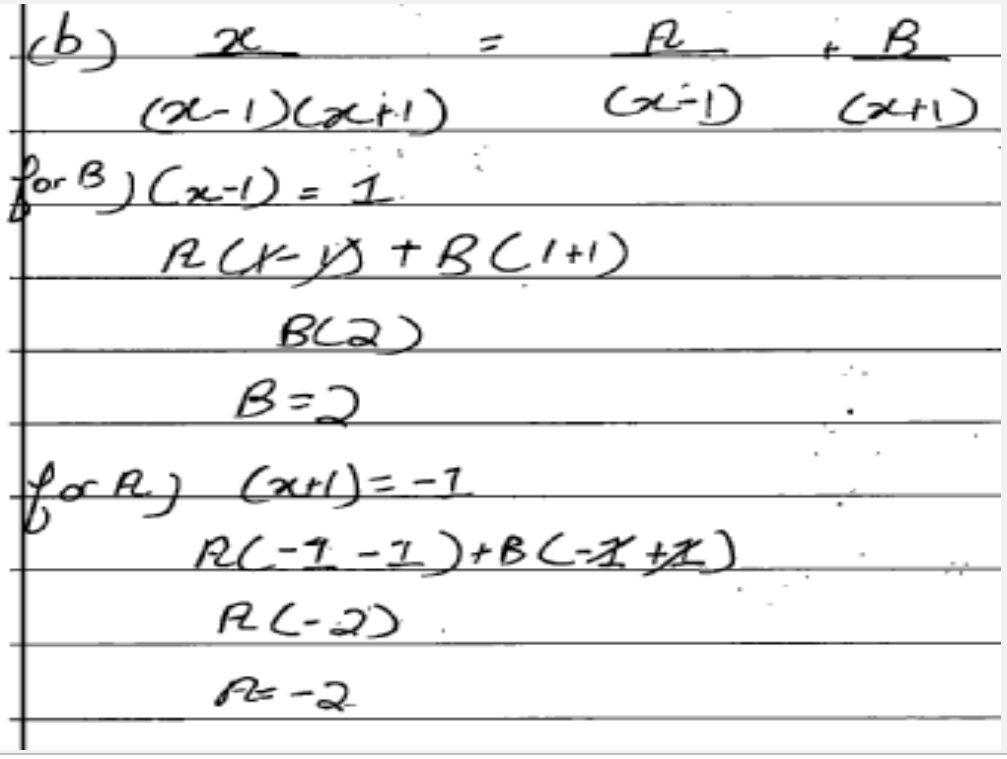
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Any Additional Suggestion:
Teachers are suggested to use the following strategies to teach the concept:
Start with concrete examples using physical objects like beads, blocks, or counters. For LCM, use multiple sets of objects and demonstrate finding the least common multiple by grouping them. For HCF, use common factors to represent shared quantities.
Use visual aids like Venn diagrams and factor trees to illustrate the relationships between numbers and how they relate to LCM and HCF. These visual representations can make the concepts more accessible.
Teach students how to perform prime factorisation. Once they can express numbers as products of prime factors, finding LCM and HCF becomes more straightforward.
Regularly review LCM and HCF concepts to reinforce learning. Use cumulative exercises that incorporate previously learned material.

Question No. 2b
Candidates were given the choice to attempt any ONE out of the two questions: 2a and 2b.

Question Text	Resolve the fraction $\frac{x}{(x-1)(x+1)}$ into partial fractions.
SLO No.	16.2.1
SLO Text	Convert an algebraic fraction into partial fractions when its denominator consists of a. non-repeated linear factors b. repeated linear factors.
Max Marks	4
Cognitive Level	A

Checking Hints	1 mark for writing in the form $\frac{x}{(x-1)(x+1)} = \frac{A}{x+1} + \frac{B}{x-1}$ 1 mark for writing in the form $x = A(x-1) + B(x+1)$ 1 mark for finding the value of A 1 mark for finding the value of B
Overall Performance	This question was related to resolving a rational fraction into its partial fractions. Most of the candidates successfully scored full marks. This part attempted by majority of the candidates.
Description of Better Responses	Better responses began with the LCM of denominators as $(x-1)(x+1)$, enabling direct fraction comparison. Then, they solved equations to find the value of A, followed by a similar process for B. This demonstrated a strong understanding of partial fractions and systematic equations.
Image of Better Response	
Description of Weaker Responses	Weaker responses indicated confusion in simplification of algebraic fractions. Some candidates were unable to calculate the LCM of the denominators, while some candidates were confused between putting $x - 1 = 0$ and $x + 1 = 0$ to get the answer and skipped the step of clearing the equation of fractions while others made minor errors that led them to the wrong answer.

Images of Weaker Responses	**Image (i)**  $\frac{x}{(x-1)(x+1)} \Rightarrow \frac{A}{(x-1)} + \frac{B}{(x+1)}$ $x = A(x-1) + B(x+1)$ $x = A(x-1=0) + B(x+1=0)$ $A = x = 1$, $B = x = -1$ $A = \frac{x}{1}$, $B = \frac{x}{-1}$ Answer **Image (ii)**  (b) $\frac{x}{(x-1)(x+1)} = \frac{A}{(x-1)} + \frac{B}{(x+1)}$ for B) $(x-1) = 1$ $A(x-y) + B(1+1)$ $B(2)$ $B = 2$ for A) $(x+1) = -1$ $A(-1-1) + B(-x+x)$ $A(-2)$ $A = -2$

Suggestions for Improvement (Highlighted part)

How to Approach SLO	Pedagogy Used for that SLO	Assessment Strategies
• Understand the expectations of the command words	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping	• Past paper questions • Discussion on E-Marking Notes

• Look at the cognitive level • Identify the content that is required to answer that question (both in terms of understanding of concepts and any skills that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Audio Visual resources • Think, Pair and Share • Questioning Technique (Socratic approach) • Practical Demonstration	• AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:
Below are some teaching strategies that can aid students in comprehending and applying partial fractions:
Visual Aids and Diagrams: Use visual aids like diagrams to illustrate the decomposition of fractions into partial fractions. Show how fractions can be broken down into simpler components.
Step-by-Step Breakdown: Provide a step-by-step breakdown of the process, starting with the basics. Begin with simple fractions and gradually move to more complex ones.
Interactive Software: Use an interactive math software that allows you to write and manipulate equations. This can help you visually demonstrate the process of decomposing fractions i.e., wolframalpha and emathhelp.net.

Question No. 3a

Candidates were given the choice to attempt any ONE out of the two questions: 3a and 3b.

Question Text	Solve the equation $\frac{\lvert x-1\rvert}{2}=1$.
SLO No.	17.2.2
SLO Text	Solve equations involving absolute value in one variable.
Max Marks	3
Cognitive Level	A
Checking Hints	1 mark for writing in the form $\lvert x-1\rvert=2$ 1 mark for writing in the form $x-1=2$ or $x-1=-2$ 1 mark for finding the values of x
Overall Performance	Majority of the candidates demonstrated a strong understanding of solving linear equations involving absolute value and correctly identified both values of x in their solutions. However, there were a few candidates who encountered difficulties. These candidates solved the absolute value equations as simple linear equations, resulting in the calculation of only one value of x. Candidates equally attempted both parts.
Description of Better Responses	Better responses exhibited correct solutions of linear equations involving absolute value and found the two values of x. Such responses excelled in handling absolute value equations. They correctly addressed the cases where the expression within the absolute value bars could be either positive or negative.

Image of Better Response	$\frac{\lvert x-1\rvert}{2} = 1$ $\lvert x-1\rvert = 2$ $x-1 = 2$ OR $x-1 = -2$ $x = 2+1$ $\quad x = -2+1$ $x = 3$ $\quad x = -1$ $S.S = \{3, -1\}$
Description of Weaker Responses	Weaker responses indicated that candidates were confused in the concept of absolute value. They were unable to differentiate between simple linear equation and absolute value linear equation. Some of the candidates solved the question like inequalities radical equations are solved which shows their lack of understanding of the concept. While some of the candidates also lost marks due to use of incorrect algebraic operations.
Images of Weaker Responses	**Image (i)** $\frac{\lvert x-1\rvert}{2} = 1$; $\frac{\lvert x+1\rvert}{2} = 1$; $\frac{\lvert x\rvert}{2} = 1-1$ $\frac{\lvert x\rvert}{2} = 1+1$; $\frac{\lvert x\rvert}{2} = 0$ $\frac{\lvert x\rvert}{2} = 2$; $\lvert x\rvert = 0$ $\lvert x\rvert = 2\times 2 = 4 = \sqrt{\lvert x\rvert^2} = \sqrt{4} = 2$; $\sqrt{\lvert x\rvert^2} = 0 = x = 0$ **Image (ii)** $\frac{\lvert x-1\rvert}{2} = 1$ $\lvert x-1\rvert = 2$ $+x-1 = 2$ $\quad -x+1 = 2$ $+x-1 > 2+1$ $\quad -x+1 > 2-1$ $x > 3$ $\quad x > 1$ $S.S = \{x \mid x \in R \wedge 3 > x > 1\}$

Suggestions for Improvement (Highlighted part)

How to Approach SLO	Pedagogy Used for that SLO	Assessment Strategies
• Understand the expectations of the command words • Look at the cognitive level • Identify the content that is required to answer that question (both in terms of understanding of concepts and any skills that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual resources • Think, Pair and Share • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:
Following are some teaching strategies related to misconceptions and common errors which candidates exhibited.
Visual Representation: Use visual aids like number lines to help students visualise the concept of absolute value. Demonstrate how it works in practice by representing numbers at different distances from zero.
Real-Life Examples: Provide real-life examples where absolute value is applicable. For instance, discuss scenarios like temperature changes or distances travelled to illustrate the concept.
Comparative Analysis: Contrast simple linear equations with absolute value equations. Show how absolute value equations typically have two solutions due to the positive and negative cases.
Misconception Addressing: Be aware of common misconceptions, such as treating absolute value equations like radical equations, and address them explicitly in your teaching. Use examples to highlight the differences.
Error Analysis: Analyse common mistakes made by students when solving absolute value equations. Encourage students to identify and correct their errors as part of the learning process.
Technology Integration: Incorporate online tools or interactive simulations that allow students to manipulate absolute value equations and visualise their solutions i.e., MathPapa and Symbollab.

Question No. 3b
Candidates were given the choice to attempt any ONE out of the two questions: 3a and 3b.

Question Text	Solve the inequality $\frac{x+2}{3} \geq 1$ and show the solution on the given number line.
SLO No.	17.4.1; 17.4.2
SLO Text	Solve linear inequalities with real coefficient, in one variable. Represent the solution of linear inequalities on the number line.
Max Marks	3

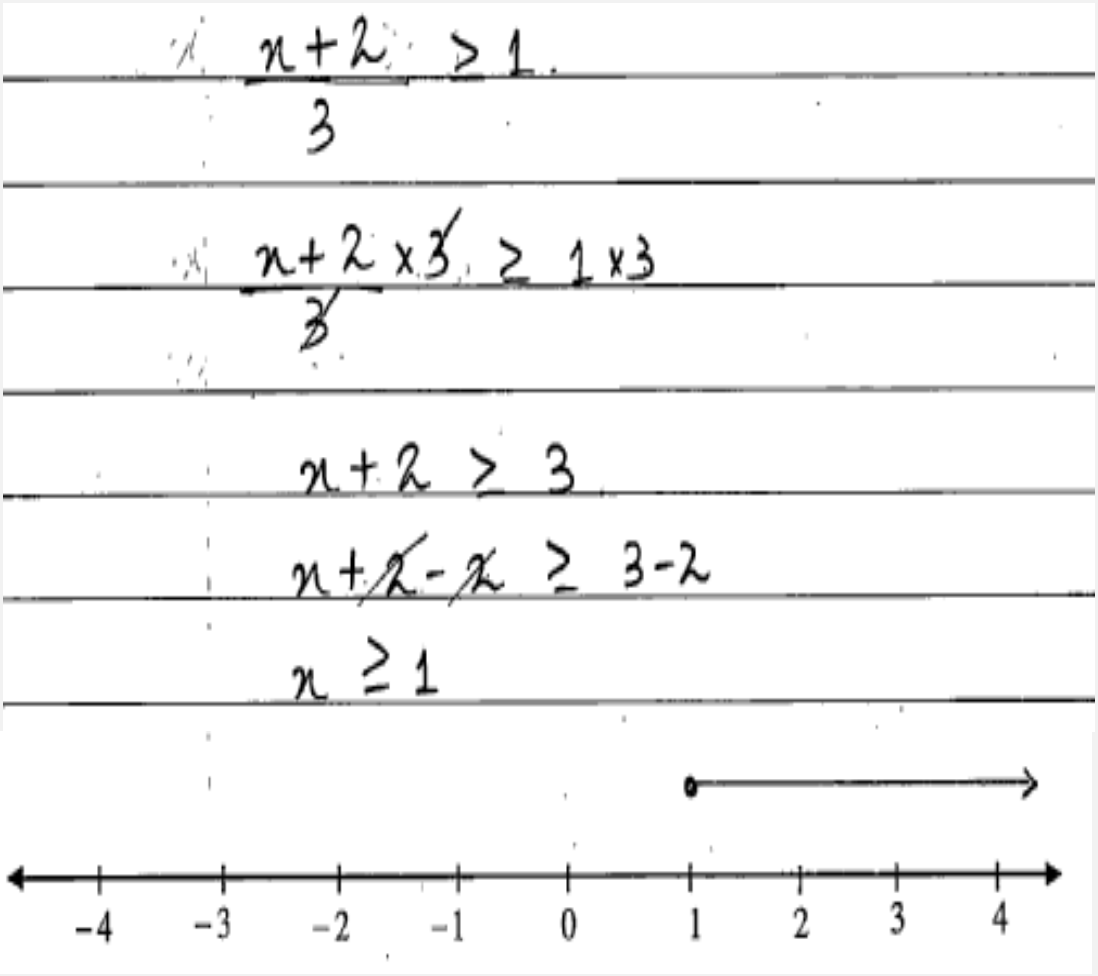
Cognitive Level	A
Checking Hints	1 mark for writing in the form of $x+2\geq 3$. 1 mark for finding the solution i.e. $x\geq 1$. 1 mark for showing the solution on the number line.
Overall Performance	This question was related to solution of linear inequalities in one variable and the representation of the solution on a number line. Overall, candidates performed relatively weaker in this question.
Description of Better Responses	Better responses not only represented the understanding of solving linear inequality but also the representation of solution on a number line. Most of the candidates showed the working of all the required step. They also drew proper number line which showed their clarity of concept.
Images of Better Responses	**Image (i)** 

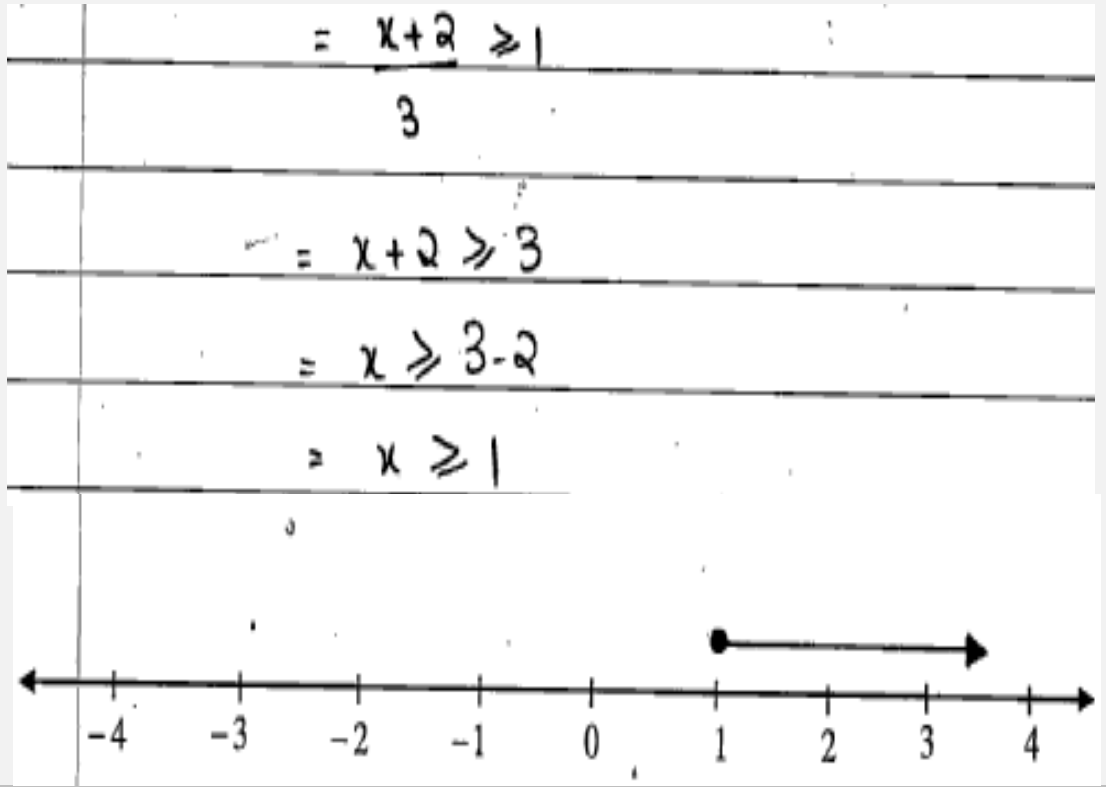
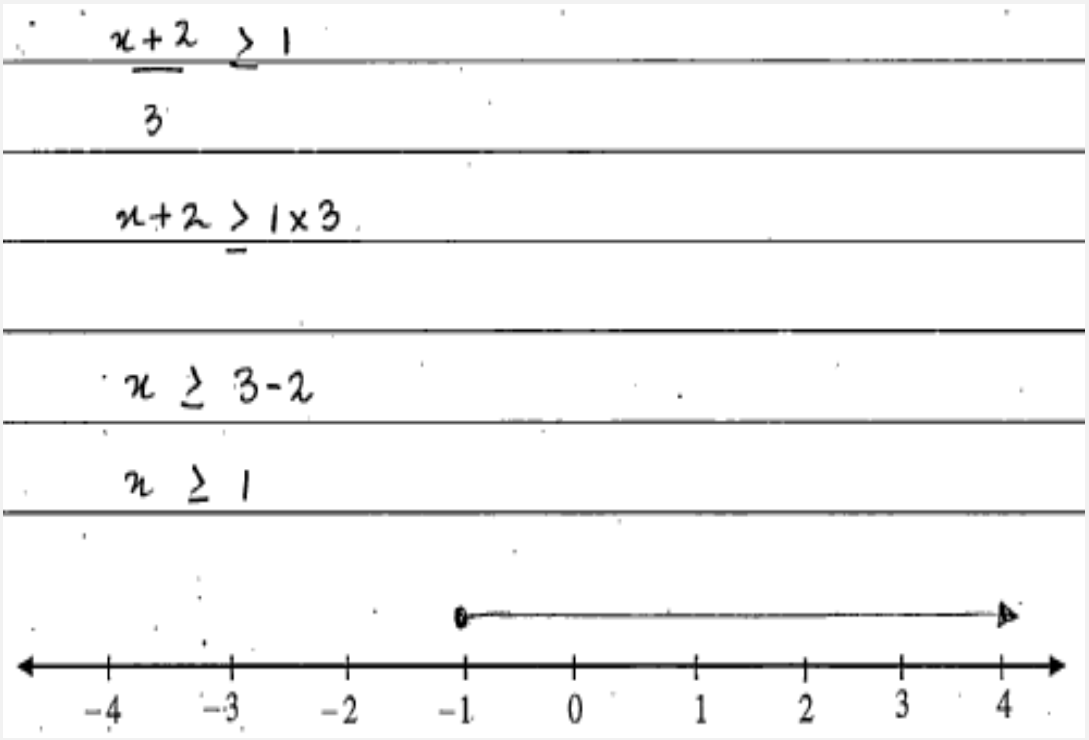
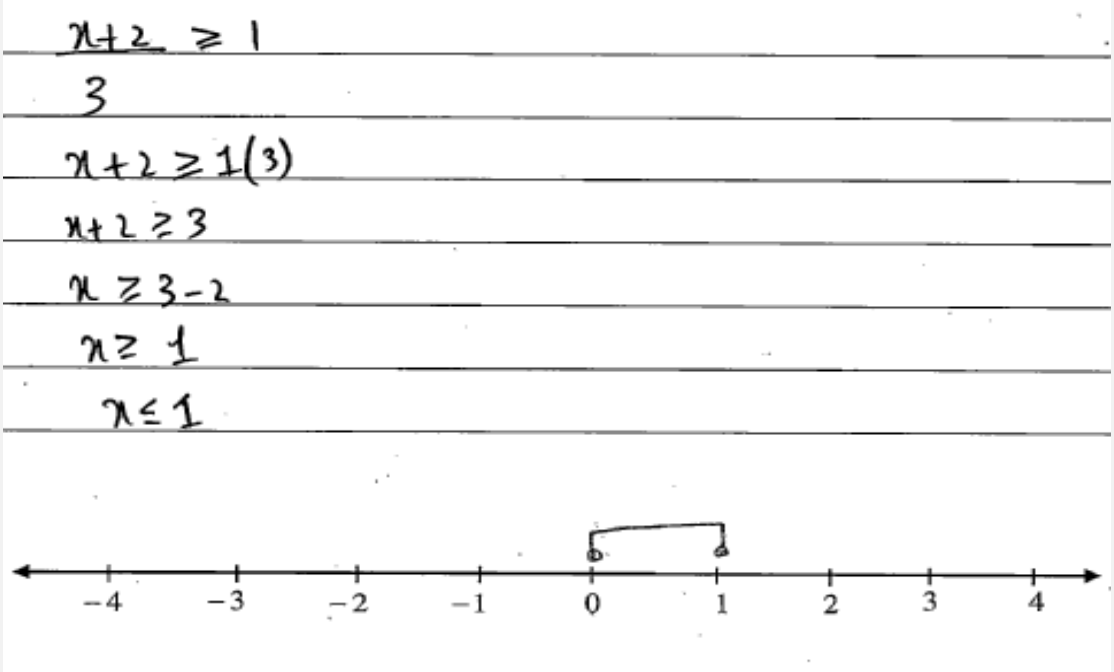
	Image (ii) 
Description of Weaker Responses	Weaker responses indicated misconception of solving in-equalities. Such responses represented confusion in the application of multiplicative property of inequality when there was a need to reverse the sign of inequality. Some candidates were unable to drew proper number line of the solution although they solved correctly. They were also confused about when to use opened and closed dots.
Images of Weaker Responses	**Image (i)** 

	Image (ii) 

Suggestions for Improvement (Highlighted part)

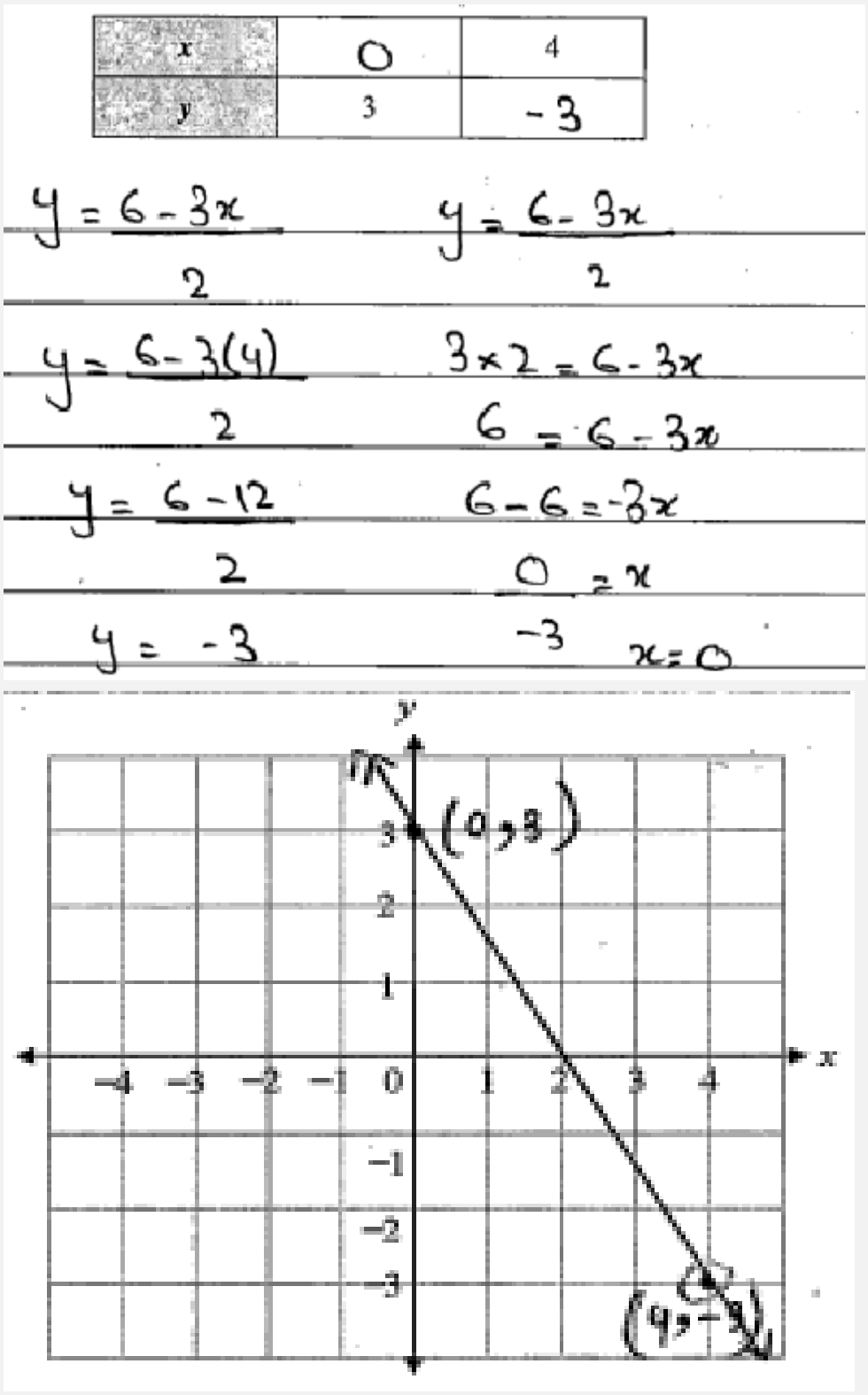
How to Approach SLO	Pedagogy Used for that SLO	Assessment Strategies
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Any Additional Suggestion:
Following are some teaching strategies related to misconceptions and common errors which candidates exhibited.
Clear Definitions and Properties: Start by providing a clear and concise explanation of inequalities, including the greater-than (>), less-than (<), greater-than-or-equal-to (≥), and less-than-or-equal-to (≤) symbols.

Visual Aids and Number Lines: Utilise visual aids, particularly number lines, to help students visualise the solutions to inequalities. Demonstrate how to represent inequalities graphically and how to determine which side of the line represents the solution set.
Interactive Activities: Engage students in interactive activities where they can manipulate number lines and inequality symbols. Online tools and interactive whiteboards can be valuable for this purpose.
Real-Life Scenarios: Present real-life scenarios that involve inequalities, such as age restrictions, budget constraints, or temperature ranges. Encourage students to translate these scenarios into mathematical inequalities.
Step-by-Step Problem Solving: Teach students a systematic approach to solving inequalities, including the application of the multiplicative property of inequalities when multiplying or dividing by negative numbers (which requires reversing the inequality sign).
Error Analysis: Analyse common mistakes made by students, particularly those related to reversing inequality signs or the use of open and closed dots on number lines. Address these errors explicitly in your teaching.
Interval Notation: Introduce interval notation as an alternative way to represent solution sets. Show students how to express solutions using brackets [ ] and parentheses ( ), and explain the difference between open and closed intervals.
Technology Integration: Use technology to create interactive simulations and quizzes that allow students to practice solving inequalities and representing solutions graphically.

Question No. 4

Question Text	i. Fill the following table for the equation $y = \frac{6-3x}{2}$. | ***x*** | | 4 | | ***y*** | 3 | | ii. Draw graph of the given equation using the values from the table in part (i).
SLO No.	18.1.6
SLO Text	Draw the graph of a given linear equation i) $y = mx$, ii) $y = mx + c$, iii) $ax + by + c = 0$
Max Marks	4
Cognitive Level	A
Checking Hints	i. 1 mark for finding value of x when y is given 1 mark for finding value of y when x is given ii. 1 mark for locating two points correctly to obtain the required graph of straight line (2 required)
Overall Performance	This question was related to identifying/ finding the points satisfying a linear equation and plotting its graph. Candidates were required to complete the table by using the values of the given variables and finding their corresponding values. Overall, candidates performed relatively weaker in this question.
Description of Better Responses	Better responses represented the understanding of solving linear equation having two variables when the value of one variable is given. These responses indicated that candidates were able to differentiate between x and y coordinates and plot the points

	carefully. Hence, they were able to obtain the straight line highlights their understating of graph of linear equation.
Image of Better Response	
Description of Weaker Responses	Weaker responses showed that candidates had difficulty in substituting the given values into equations, often used incorrect values and made simplification errors. Furthermore, such candidates were also unable when plotting points based on the calculated coordinates.

Images of Weaker Responses

Image (i)

x	-2	4
y	3	2

$y = \frac{6 - 3x}{2}$

$2y = 6 - 3x$

$2y = 6 - 3(2)$

$2y = 6 - 6$

$2y = 0$

$y = 2^{1}/_{2}$

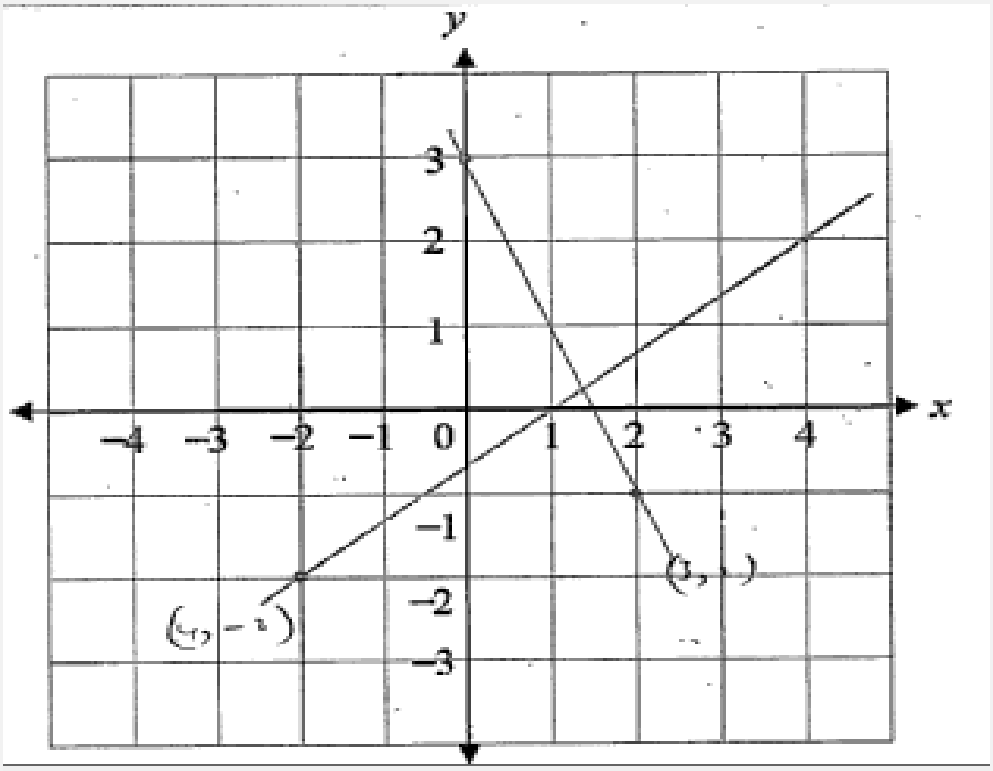


Image (ii)

x	−3	4
y	3	2

$-3 = \frac{6-3x}{2}$

$-6 = 6 - 3x$

$-6 - 6 = -3x$

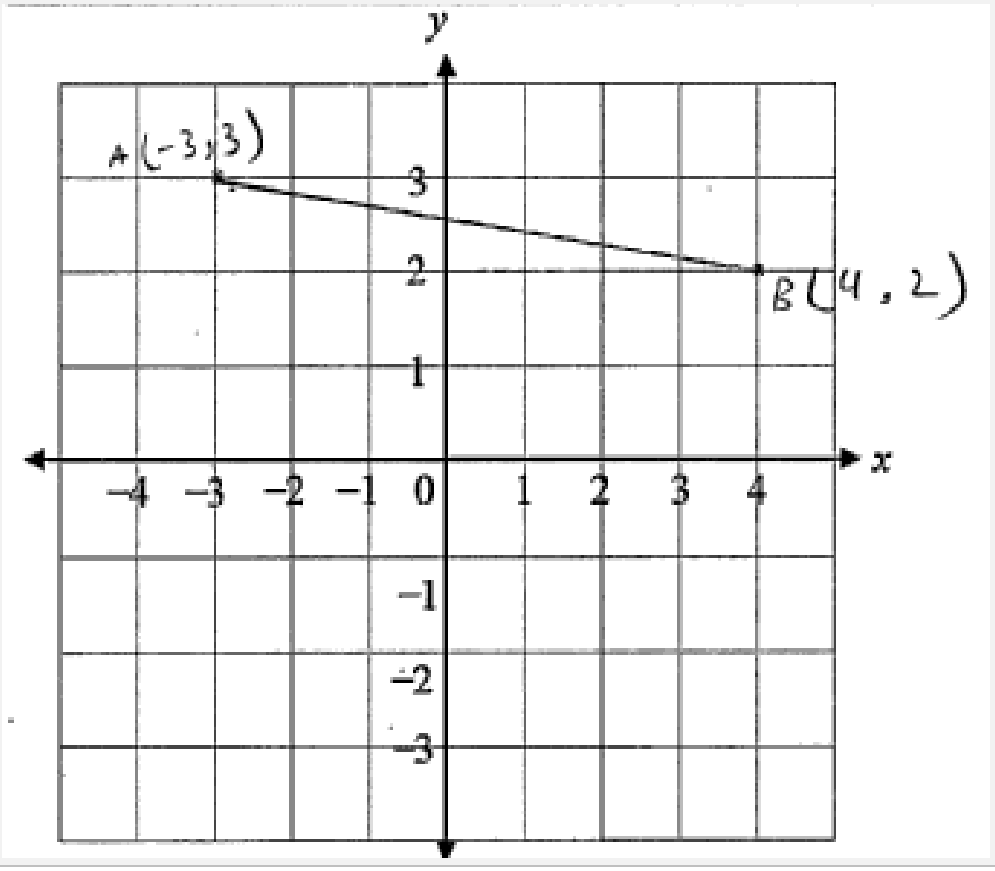
$\frac{-12}{3} = x$

$x = -4$

$y = \frac{6-3(2)}{2}$

$y = \frac{6-6}{2}$

$y = 2$

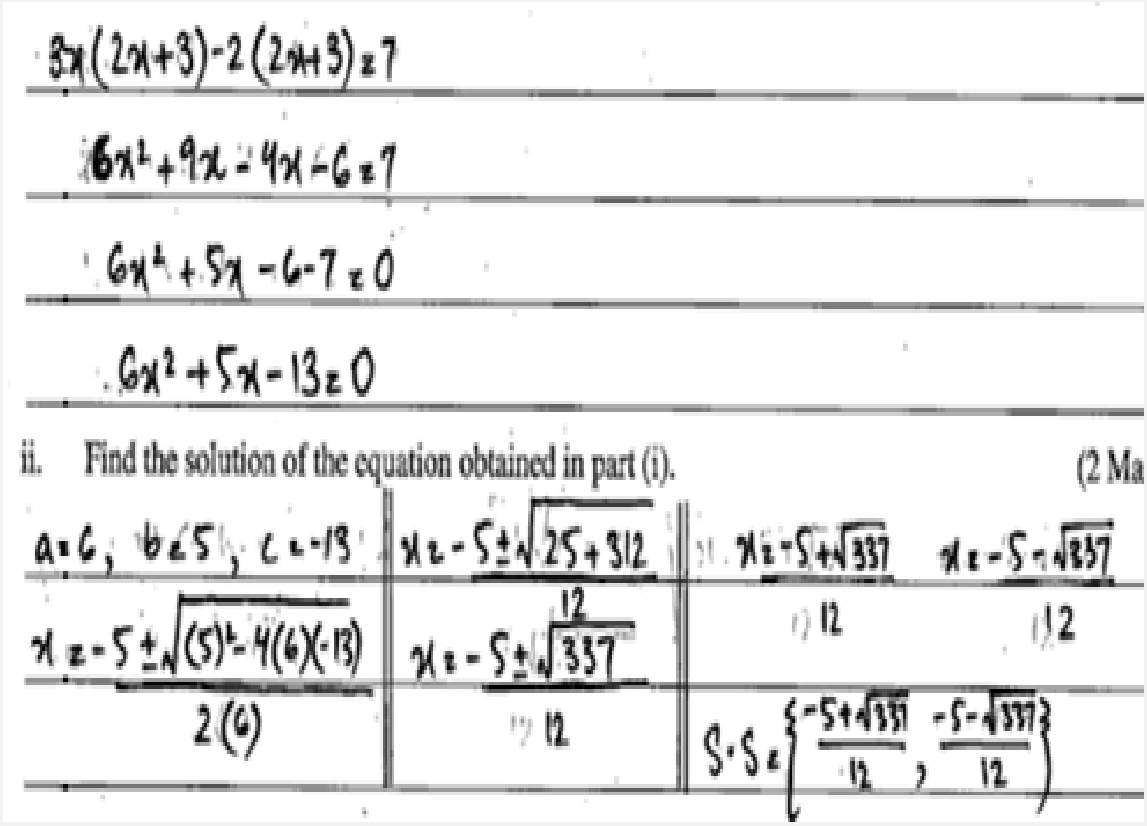


Suggestions for Improvement (Highlighted part)

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Any Additional Suggestion:
Teachers may these teaching methodologies to help students overcome the challenges associated with substituting values into equations and accurately plotting points on the graph of linear functions.
Use of Technology: Incorporate graphing calculators or graphing software to visualise linear functions. This allows students to see the graph as they make changes to the equation's parameters.
Graphing Activities: Conduct graphing activities where students plot points on a coordinate plane based on calculated coordinates. Start with basic linear equations and then progress to more challenging equations.
Visual Aids and Manipulatives: Use visual aids like graph paper, coordinate grids, and manipulatives (e.g., geometric shapes) to help students physically represent linear equations and their solutions.
Coordinate Plane Exploration: Allow students to explore the coordinate plane and practice plotting points before introducing equations. This familiarity can help them connect the algebraic and graphical aspects of linear functions.
Modeling and Application: Show how linear functions and graphs are used in real-life scenarios, such as linear growth, cost functions, or rate problems. Relate these applications to the equations and graphs studied in class.

Question No. 5	
Question Text	A quadratic equation is given as $(3x-2)(2x+3)=7$. i. Convert the equation in the form $ax^2+bx+c=0$. ii. Find the solution of the equation obtained in part (i).
SLO No.	19.3.4
SLO Text	Solve the quadratic equations involving real and complex roots using quadratic formula.
Max Marks	3
Cognitive Level	A

Checking Hints	i. 1 mark for getting quadratic equation into its standard form ii. 1 mark for the application of the quadratic formula 1 mark for the simplification to get $x = \frac{-5+\sqrt{337}}{12}$ or $x = \frac{-5-\sqrt{337}}{12}$
Overall Performance	This question was related to the conversion of quadratic equations standard form and its solution using quadratic formula. The major error that the candidates made was in conversion into standard form. Moreover, some candidates lost their marks in error while simplifying the expression.
Description of Better Responses	Better responses expressed the given equation in standard form, stated the correct quadratic formula, identified the correct values of *a*, *b* and *c*, and solved to get the correct values of *x*. While some candidates also applied the completing square method and got the required result.
Image of Better Response	
Description of Weaker Responses	Weaker responses showed that some of the candidates were unable to find the product of two factors and that was the reason they lost their marks. Some made mistakes in application of quadratic formula and some lost their marks because of error in using arithmetic operations. The major error that the candidates made was in conversion of quadratic equation into standard form.

Images of Weaker Responses

Image (i)

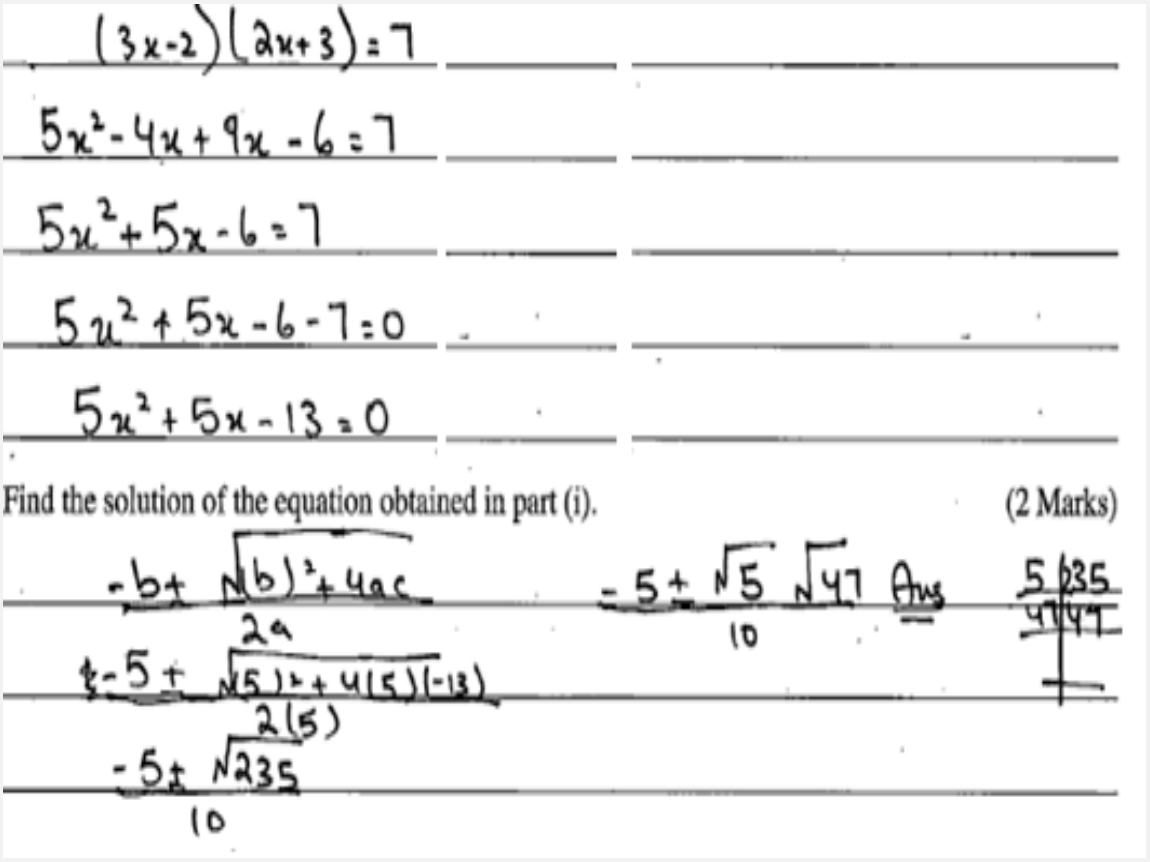
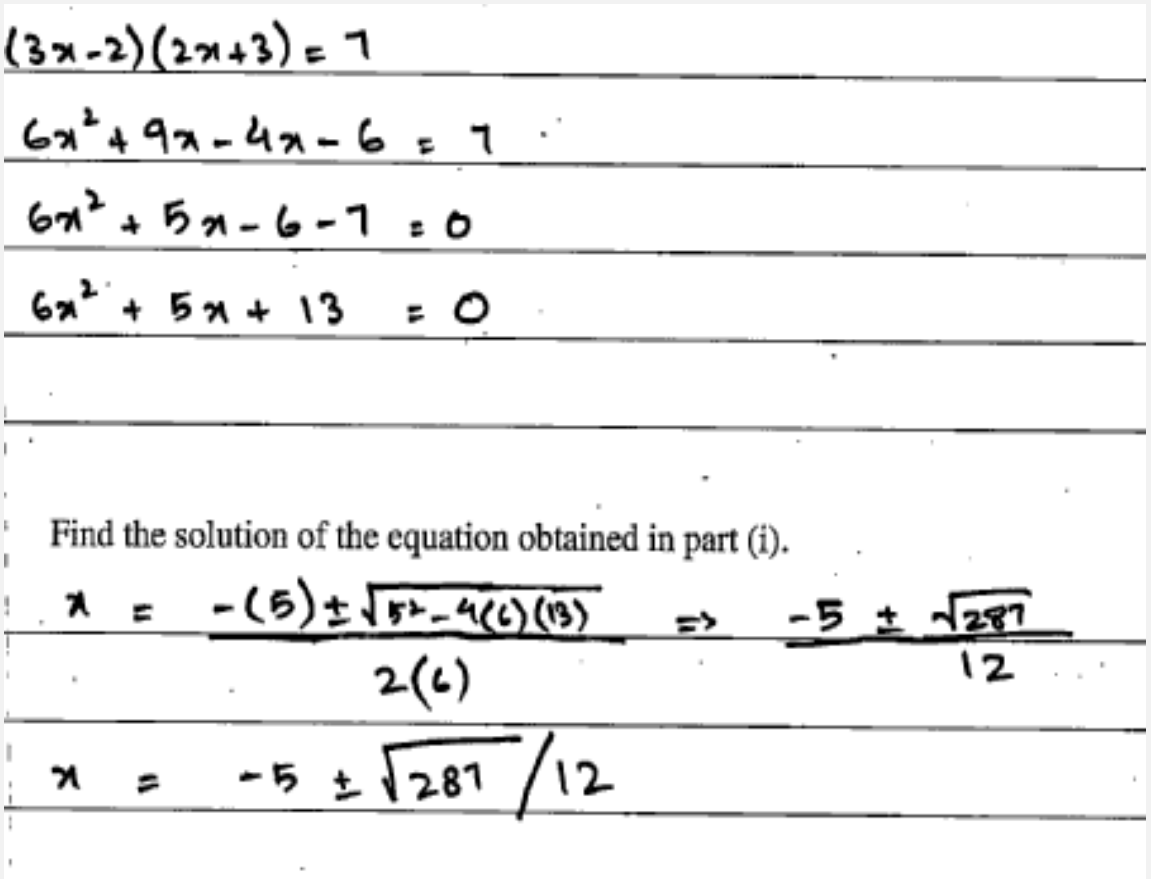


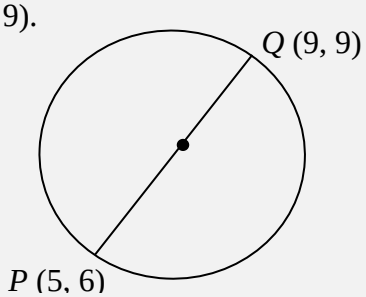
Image (ii)



Suggestions for Improvement (Highlighted part)

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Any Additional Suggestion:
Following are the teaching strategies that can be employed by teachers to successfully instruct students in solving quadratic equations and the use of the quadratic formula.
Step-by-Step Problem Solving: Break down the quadratic formula into its components (discriminant, coefficients, etc.), and guide students through each step of solving a quadratic equation. Encourage them to ask questions and seek clarification as needed.
Hands-On Manipulatives: Use algebra tiles or physical manipulatives to model quadratic equations. This tactile approach can help students visualise the equations and roots.
Error Analysis: Share incorrect solutions to quadratic equations and ask students to identify and correct errors. This helps them develop a critical eye and understand common mistakes.
Conceptual Discussions: Engage in class discussions about the nature of roots, how they relate to the coefficients, and the significance of the discriminant in determining the types of roots.

Question No. 6	
Question Text	In the diagram, a circle with centre O is given. The end points of the diameter of the circle are P (5, 6) and Q (9, 9).  **NOT TO SCALE** Find the coordinates of the point O and the radius of the circle.
SLO No.	20.1.3 and 20.3.2

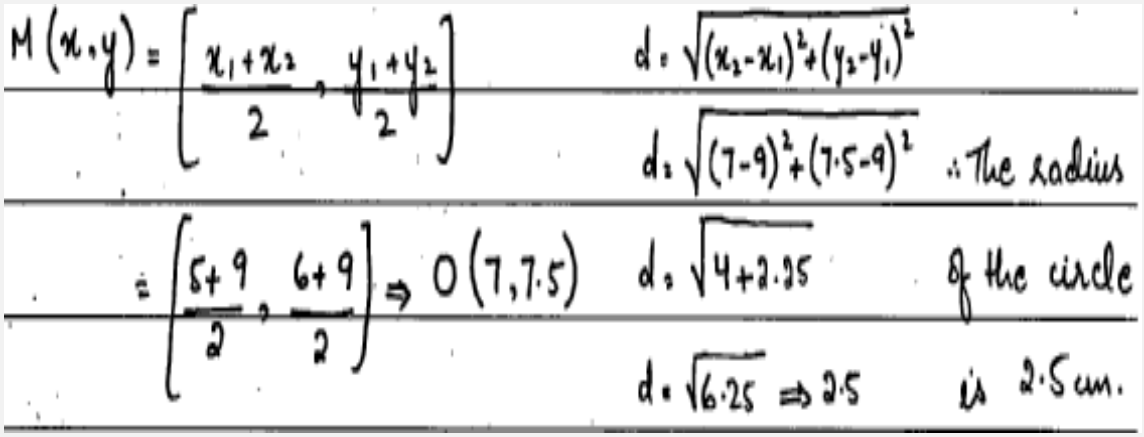
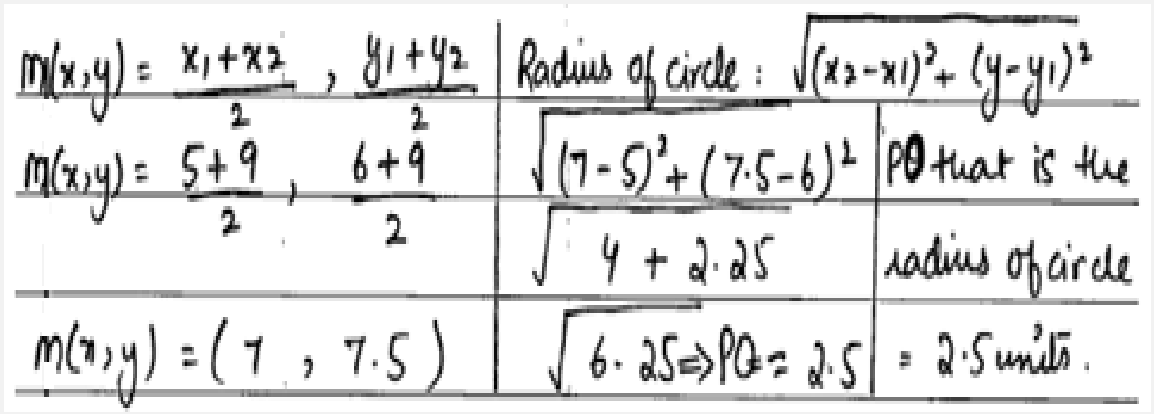
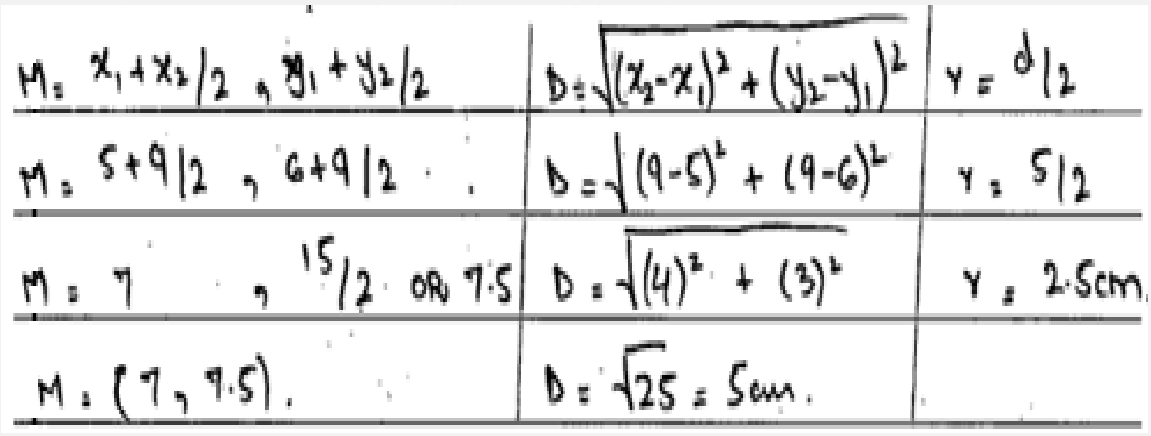
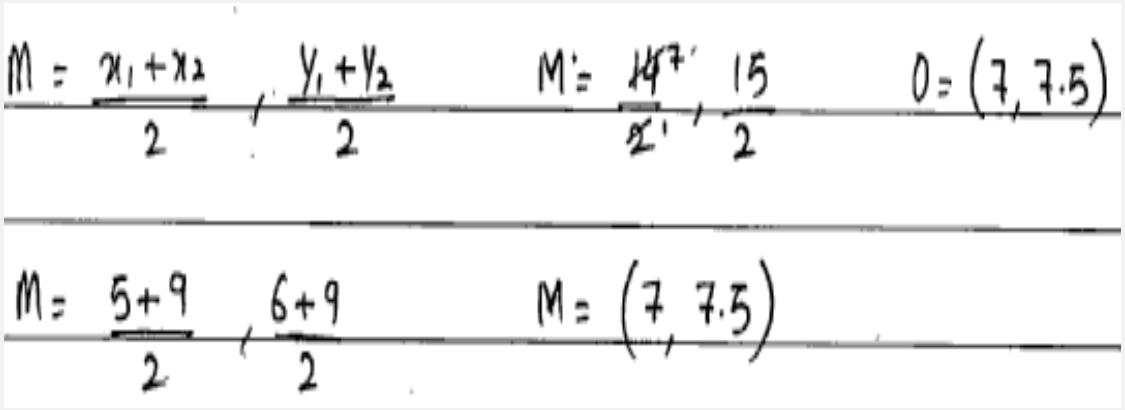
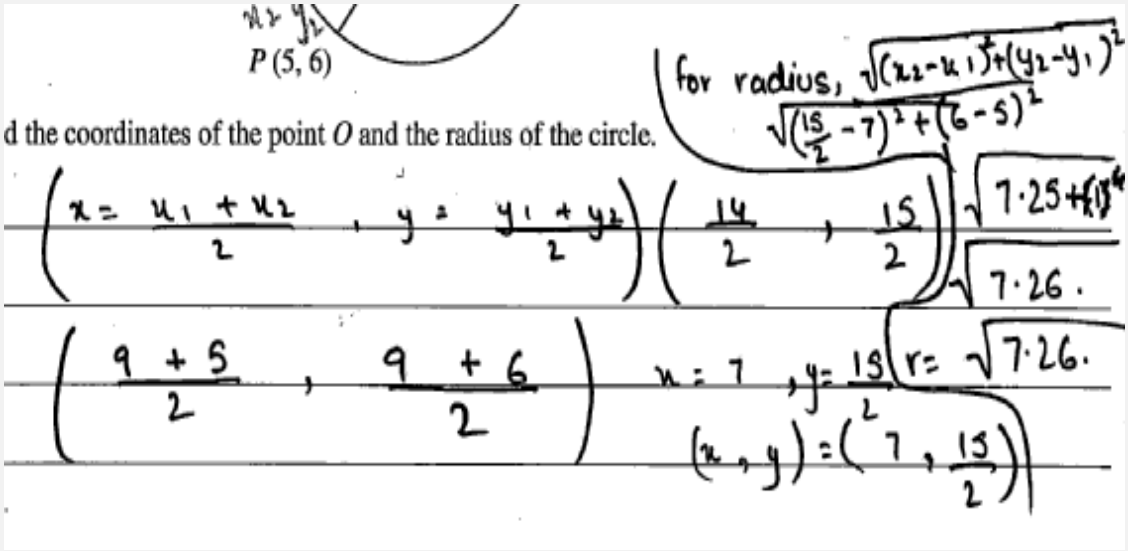
SLO Text	Find the distance between two given points using distance formula. Solve problems related to midpoint formula.
Max Marks	3
Cognitive Level	A
Checking Hints	1 mark for finding the centre of the circle 1 mark for substituting the values in the distance formula, i.e. $\frac{1}{2}\sqrt{(9-5)^2+(9-6)^2}$ 1 mark for the simplification to get the radius of the circle
Overall Performance	This question was related to the concept of midpoint distance formula and its application. Candidates were required to find the midpoint of the diameter, that is the centre of the circle. In addition, they were required to find the distance from the centre to any of the two points on the circumference as the radius. However, the common errors include application of incorrect formula and incorrect substitution of points.
Description of Better Responses	Better responses used the midpoint formula to find the coordinates of centre of the circle. They applied the formula correctly to get the required results. Also indicated different methods to calculate the radius of the circle by using distance formula. Some of the candidates first found diameter *PQ* and then, divided by 2 to find radius and some calculated radius *PO* or *OQ* directly with the help of distance formula.
Images of Better Responses	**Image (i)**  $M(x,y) = \left[\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right]$ $= \left[\frac{5+9}{2}, \frac{6+9}{2}\right] \Rightarrow O(7,7.5)$ $d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ $d = \sqrt{(7-9)^2+(7.5-9)^2}$ $d = \sqrt{4+2.25}$ $d = \sqrt{6.25} \Rightarrow 2.5$ ∴ The radius of the circle is 2.5 cm. **Image (ii)**  $m(x,y) = \frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}$ $m(x,y) = \frac{5+9}{2}, \frac{6+9}{2}$ $m(x,y) = (7, 7.5)$ Radius of circle : $\sqrt{(x_2-x_1)^2+(y-y_1)^2}$ $\sqrt{(7-5)^2+(7.5-6)^2}$ $\sqrt{4+2.25}$ $\sqrt{6.25} \Rightarrow PO = 2.5$ PO that is the radius of circle = 2.5 units.

	Image (iii) 
Description of Weaker Responses	Weaker responses demonstrated that some candidates only calculated the coordinates of centre of the circle by using mid-point formula. While others used the distance formula to find the radius; however, they substituted the incorrect coordinates.
Images of Weaker Responses	**Image (i)**  **Image (ii)** 

Suggestions for Improvement (Highlighted part)

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Any Additional Suggestion:

Teachers are encouraged to employ these interactive teaching strategies to enhance student engagement and comprehension:

Visual Representation: Begin by visually representing the distance formula using a coordinate plane. Use two points and draw the right triangle that represents the distance between them. This helps students see the geometric interpretation.

Class Collaborative Activity: Organise a class activity where students measure the distance between various points in the classroom or school using measuring tapes or rulers. Discuss how this relates to the formula.

Real-Life Applications: Show real-life scenarios where the distance formula is used, such as calculating the distance between two locations on a map, finding the length of a rope between two points, or determining the range of a wireless signal.

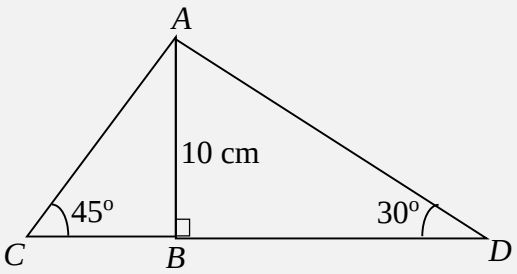
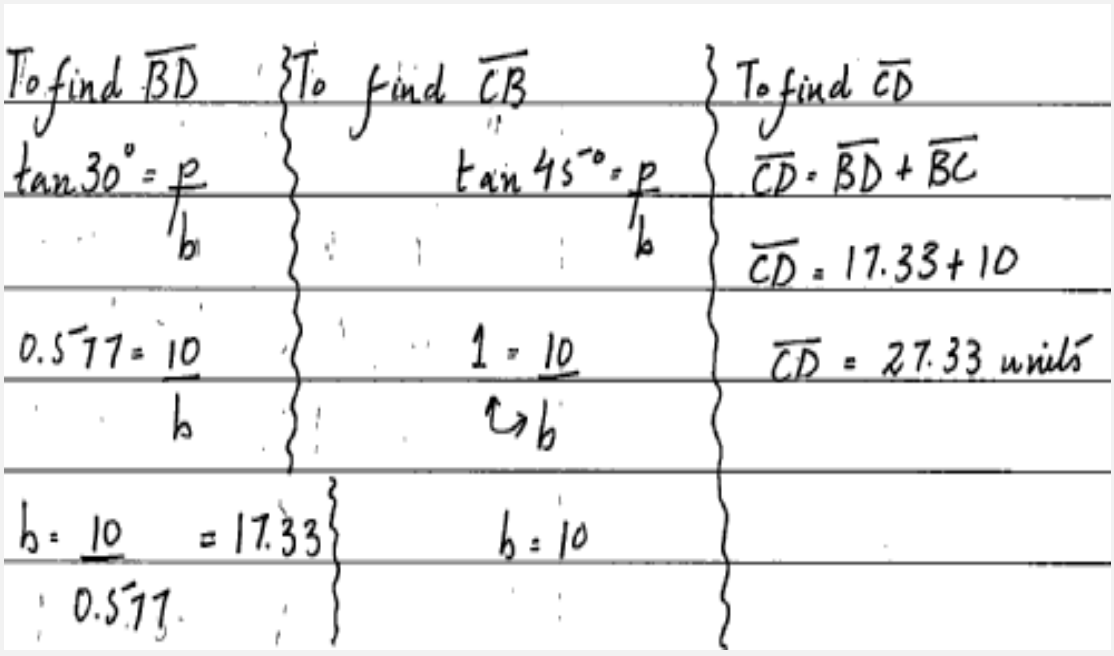
Geometric Visualisation: Start with a visual representation of the midpoint formula. Use a coordinate plane and two points to illustrate how the midpoint is located at the centre of the line segment.

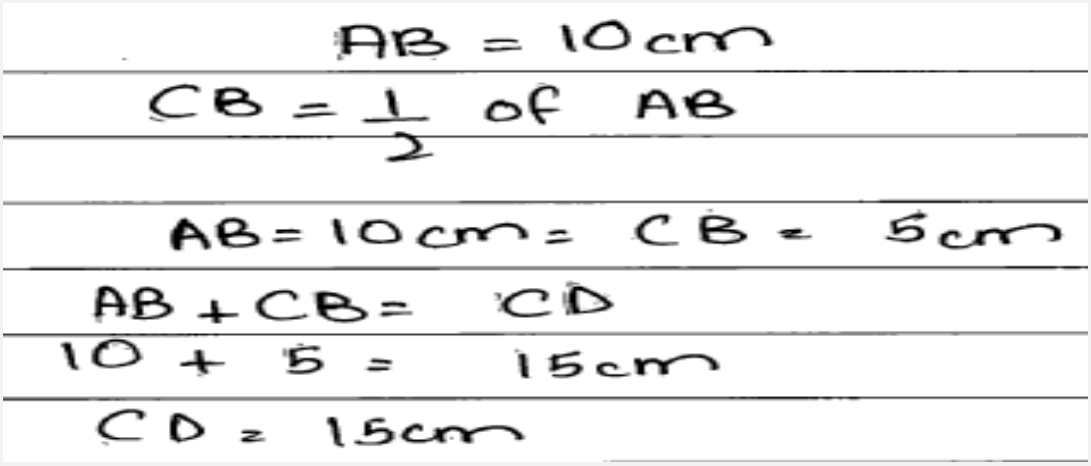
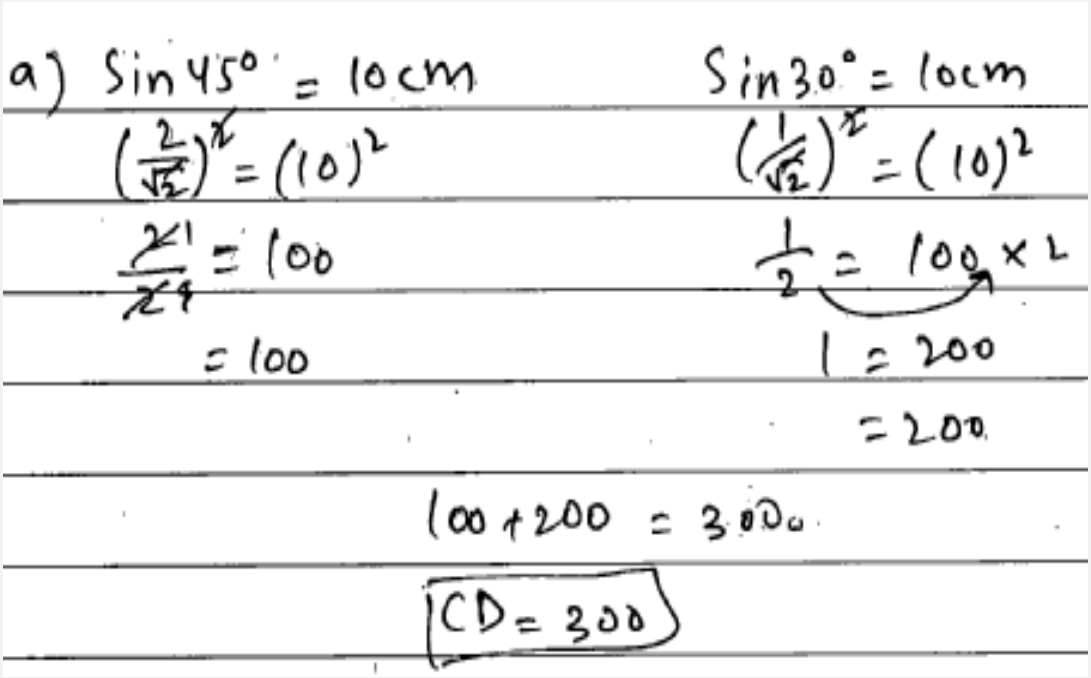
Concrete Manipulatives: Provide physical manipulatives like string or rope and ask students to find the midpoint of different segments. This hands-on approach reinforces the concept.

Real-Life Scenarios: Share examples where the midpoint formula is used in real-life situations, such as dividing a line segment in a sports field, locating the centre of a painting, or finding the balance point of an object.

Question No. 7a

Candidates were given the choice to attempt any ONE out of the two questions: 7a and 7b.

Question Text	In the given diagram, find the length of *CD*.  **NOT TO SCALE**
SLO No.	21.5.2
SLO Text	Solve problems involving angle of elevation and depression in a right angled triangle.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark applying correct trigonometric ratios 1 mark for finding the value of *BC* 1 mark for finding the value of *BD* 1 mark for finding the value of *CD*
Overall Performance	A majority of candidates faced difficulties in their attempts, primarily stemming from the application of incorrect ratios in their calculations. Additionally, some candidates resorted to Pythagoras theorem to find lengths, introducing complexity to their solutions.
Description of Better Responses	Better responses applied properties of right-angled triangle to solve the question successfully which indicates their critical thinking. They used the correct formula of tangent to find the values of $\overline{BC}$ and $\overline{BD}$ and added the lengths to get the final answer.
Image of Better Response	
Description of Weaker Responses	Weaker responses represented misunderstandings of the application of trigonometric ratios. Some candidates were unable to select the correct trigonometric ratio or substitute the correct value in the ratio to find the value of $\overline{BC}$ and $\overline{CD}$.

Images of Weaker Responses	**Image (i)**  AB = 10 cm CB = 1/2 of AB AB = 10 cm = CB = 5 cm AB + CB = CD 10 + 5 = 15 cm CD = 15 cm **Image (ii)**  a) Sin 45° = 10 cm $(\frac{2}{\sqrt{2}})^x = (10)^2$ $\frac{x^1}{x^1} = 100$ = 100 Sin 30° = 10 cm $(\frac{1}{\sqrt{2}})^x = (10)^2$ $\frac{1}{2} = 100 \times 2$ 1 = 200 = 200 100 + 200 = 300 CD = 300

Suggestions for Improvement (Highlighted part)

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concepts and any skills that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Questioning Technique (Socratic approach) • Practical Demonstration	

Any Additional Suggestion:

Teachers are suggested to use following teaching strategies and real-life scenarios, that can help students grasp the concept of angle of elevation and angle of depression in a right-angled triangle effectively:

Begin with a visual demonstration using a projector or interactive whiteboard. Show pictures or animations illustrating angle of elevation and angle of depression scenarios, such as a person looking up at a tree (elevation) or down at a boat (depression).

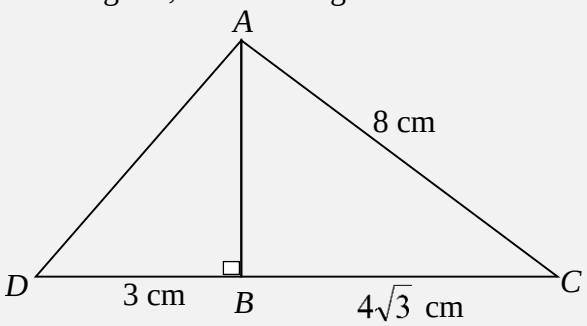
Provide students with real-life scenarios that involve angle of elevation and angle of depression. For example, describe situations where someone is measuring the height of a building or the depth of a pool.

Take students outdoors for an observation activity. Find objects or structures where they can estimate angles of elevation and depression. Let them measure and record angles using protractors.

Provide physical props and models like a toy boat, a tree, and a measuring tape. Ask students to create their own angle of elevation and depression scenarios and measure the angles involved.

Question No. 7b

Candidates were given the choice to attempt any ONE out of the two questions: 7a and 7b.

Question Text	In the given diagram, find the length of AD.  **NOT TO SCALE**
SLO No.	22.1.1
SLO Text	Students will be able to apply the following theorems. i. in a right-angled triangle, the square of the length to hypotenuse is equal to the sum of the squares of the lengths of the other two sides (Pythagoras' theorem). ii. if the square of one side of a triangle is equal to the sum of the squares of the other two sides then the triangle is a right angled triangle.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark applying correct trigonometric ratios 1 mark for finding the value of BC 1 mark for finding the value of BD 1 mark for finding the value of CD

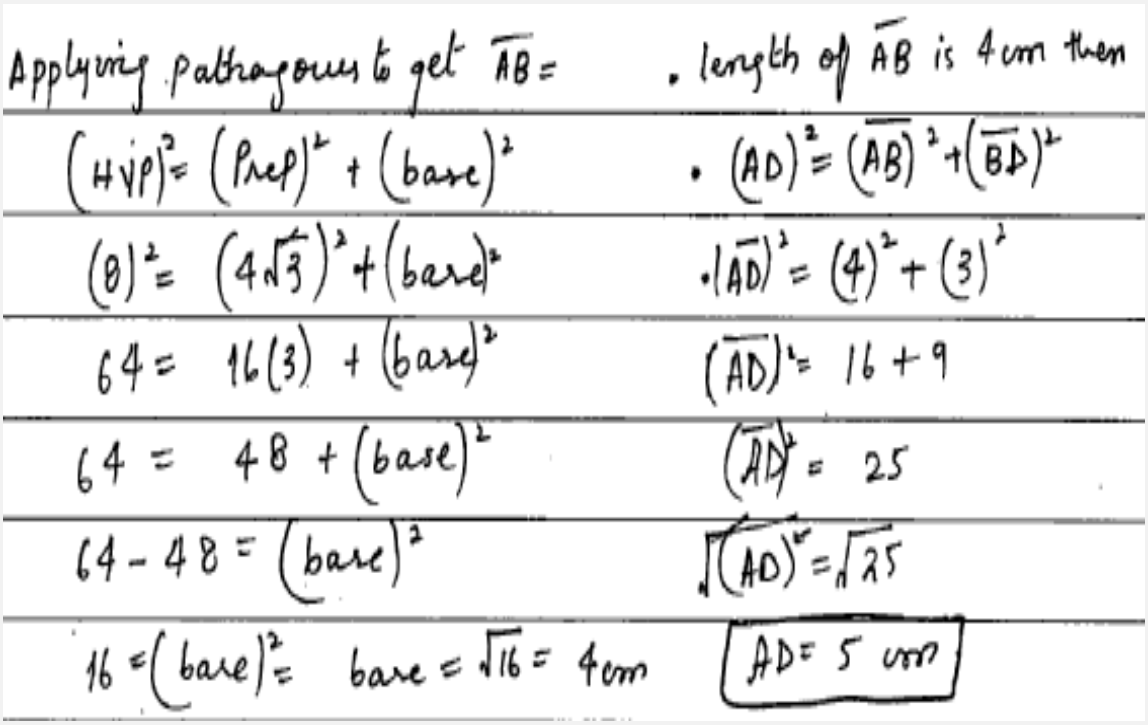
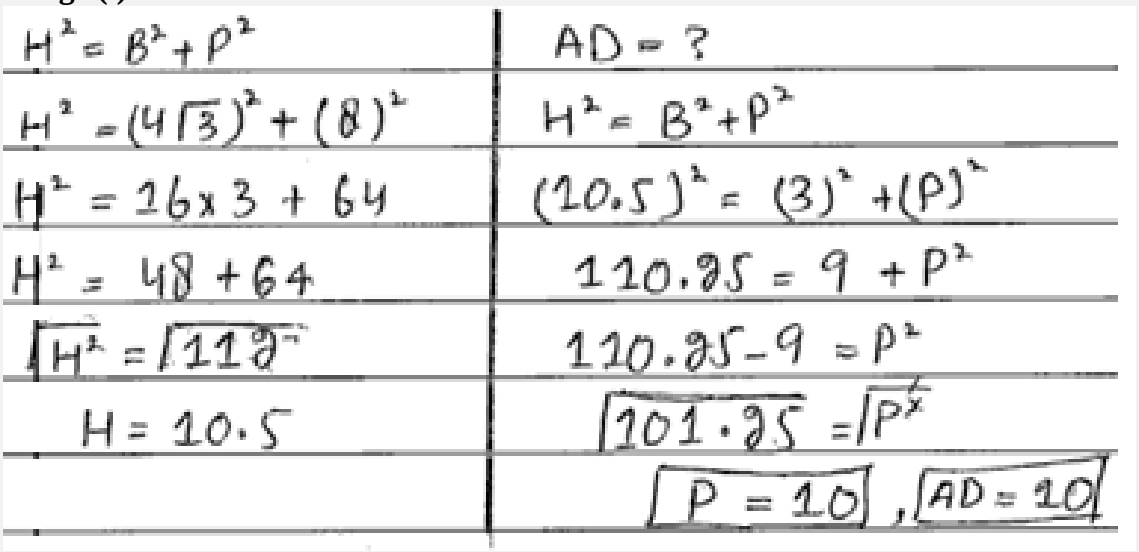
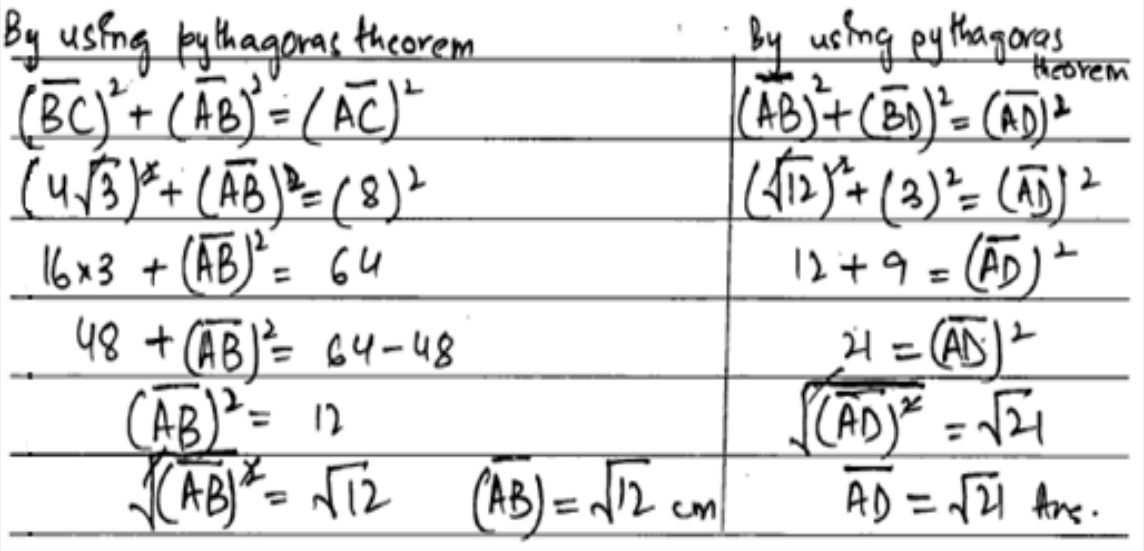
Overall Performance	Question 7b required the candidates to find the unknown sides of the given right angled triangles using Pythagoras theorem. Overall, candidates performed well in this question. This part attempted by majority of the candidates.
Description of Better Responses	Better responses showed that candidates were capable of solving particular triangle with the help of Pythagoras theorem. Their working showed that they understood that to find the length of *AD*, first they had to find the length of *AB* by using Pythagoras theorem. They were also able to find the square of irrational number. These candidates did all the required steps for the particular question so they were awarded full marks
Image of Better Response	
Description of Weaker Responses	Weaker responses indicated that candidates knew basic definition of the Pythagoras theorem; however, there was confusion in its application. Some candidates were unable to identify which side is hypotenuse. The other common mistake which most of the candidates made was finding the square of $4\sqrt{3}$. Some of the candidates also made mistakes in calculations and lost their marks.
Images of Weaker Responses	**Image (i)** 

Image (ii)



Suggestions for Improvement (Highlighted part)

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Any Additional Suggestion:

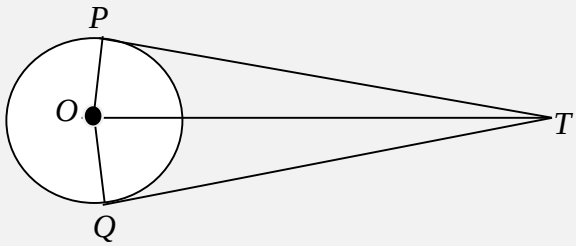
While teaching the concept of Pythagoras theorem, teachers are advised to use the following strategies:

Start with real-life examples where the Pythagorean Theorem is used. For instance, explain how architects and builders use it to ensure the stability and accuracy of structures. Show how it's applied in navigation, such as calculating distances on maps or GPS systems.

Use interactive demonstrations with props like wooden blocks to physically show the concept of the theorem. Create right-angled triangles with sides of known lengths and ask students to verify the theorem by measuring the sides and calculating the square of the hypotenuse.

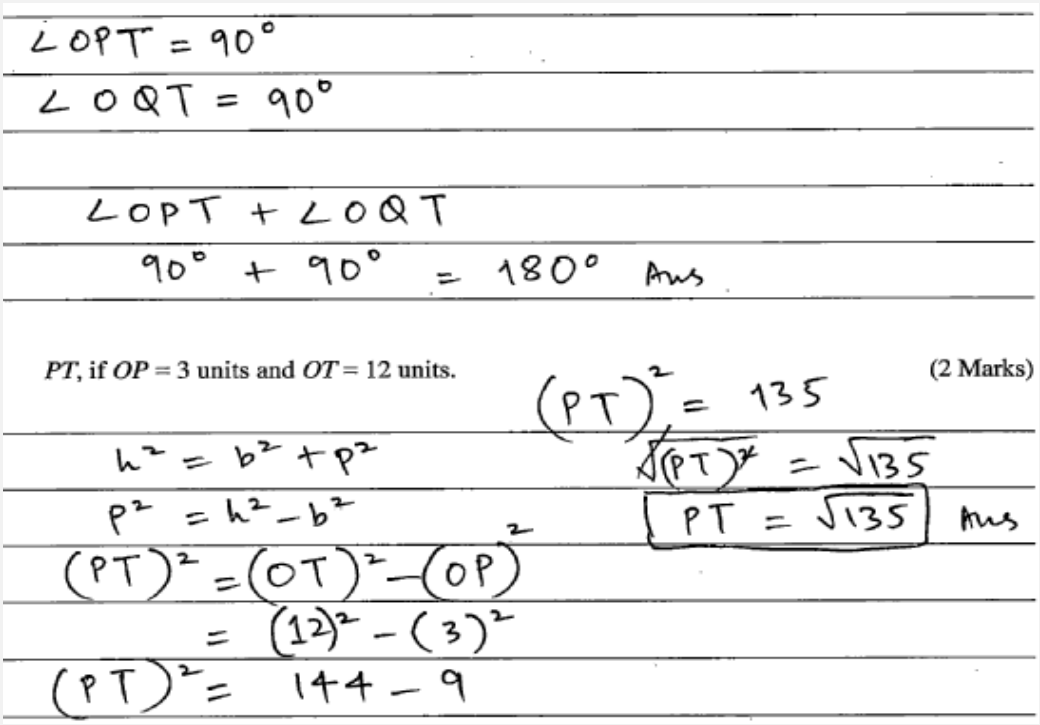
Conduct hands-on activities where students construct right-angled triangles using materials like string, rulers, and protractors. Have them measure the sides and calculate the hypotenuse to reinforce the theorem.

Use quizzes, worksheets, or interactive online platforms to assess students' understanding. Provide timely feedback to address misconceptions and reinforce learning.

Question No. 8a Candidates were given the choice to attempt any ONE out of the two questions: 8a and 8b.	
Question Text	In the given diagram, *O* is the centre of the circle and *PT* and *QT* are the two tangents drawn to the circle.  **NOT TO SCALE** Find the value of i. $\angle OPT + \angle OQT$. ii. *PT*, if *OP* = 3 units and *OT* = 12 units.
SLO No.	24.1.1b and 24.1.1c
SLO Text	Apply the following theorems to solve related problems. b. the tangent to a circle and the radial segment joining the point of contact and the centre are perpendicular to each other. c. the two tangents drawn to a circle from a point outside it, are equal in length.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark application of Pythagorean Theorem on triangle *ABC* 1 mark for finding the value of *AB* 1 mark application of Pythagorean Theorem on triangle *ABD* 1 mark for finding the value of *AD*
Overall Performance	In part (i) of the question, only a limited number of candidates were able to recall and accurately apply the theorem stating that "the tangents drawn to a circle from a point outside a circle are equal in measure." A majority of the candidates faced difficulties, particularly in identifying the angle between the outer point of the radial segment and the tangent at that point. Moving on to part (ii), a different pattern emerged. Here, a majority of the candidates were able to successfully apply the Pythagoras theorem. However, it's worth noting that many of these candidates substituted incorrect values for the hypotenuse, resulting in errors in their calculations.
Description of Better Responses	Better responses stated the correct values and got full marks. For part (i), $\angle OPT + \angle OQT = 90^{o} + 90^{o} = 180$, although in the question reason was not required; however, some candidates mentioned the reason too which showed their understanding of the topic. For part (ii), candidates applied Pythagorean theorem to find the length of *AD*.

Images of Better Responses

Image (i)



$\angle OPT = 90^\circ$

$\angle OQT = 90^\circ$

$\angle OPT + \angle OQT$

$90^\circ + 90^\circ = 180^\circ$ Ans

PT, if *OP* = 3 units and *OT* = 12 units. (2 Marks)

$h^2 = b^2 + p^2$

$p^2 = h^2 - b^2$

$(PT)^2 = (OT)^2 - (OP)^2$

$= (12)^2 - (3)^2$

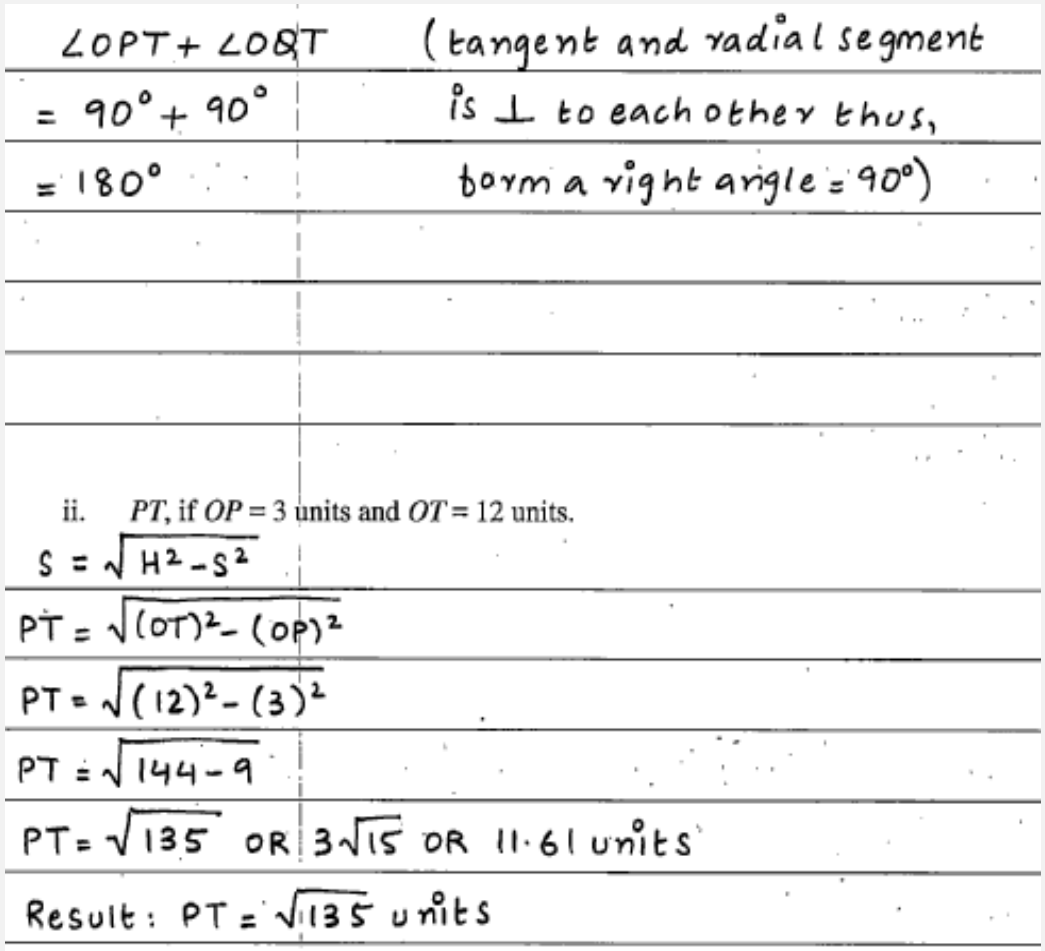
$(PT)^2 = 144 - 9$

$(PT)^2 = 135$

$\sqrt{(PT)^2} = \sqrt{135}$

$PT = \sqrt{135}$ Ans

Image (ii)



$\angle OPT + \angle OQT$ (tangent and radial segment

$= 90^\circ + 90^\circ$ is ⊥ to each other thus,

$= 180^\circ$ form a right angle = 90°)

ii. *PT*, if *OP* = 3 units and *OT* = 12 units.

$S = \sqrt{H^2 - S^2}$

$PT = \sqrt{(OT)^2 - (OP)^2}$

$PT = \sqrt{(12)^2 - (3)^2}$

$PT = \sqrt{144 - 9}$

$PT = \sqrt{135}$ OR $3\sqrt{15}$ OR 11.61 units

Result: $PT = \sqrt{135}$ units

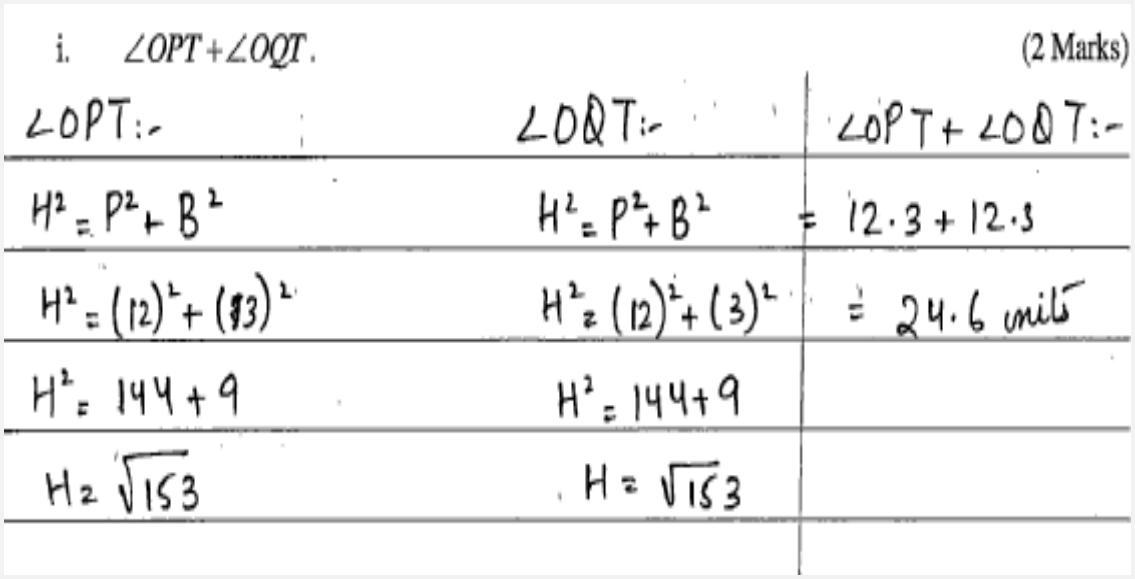
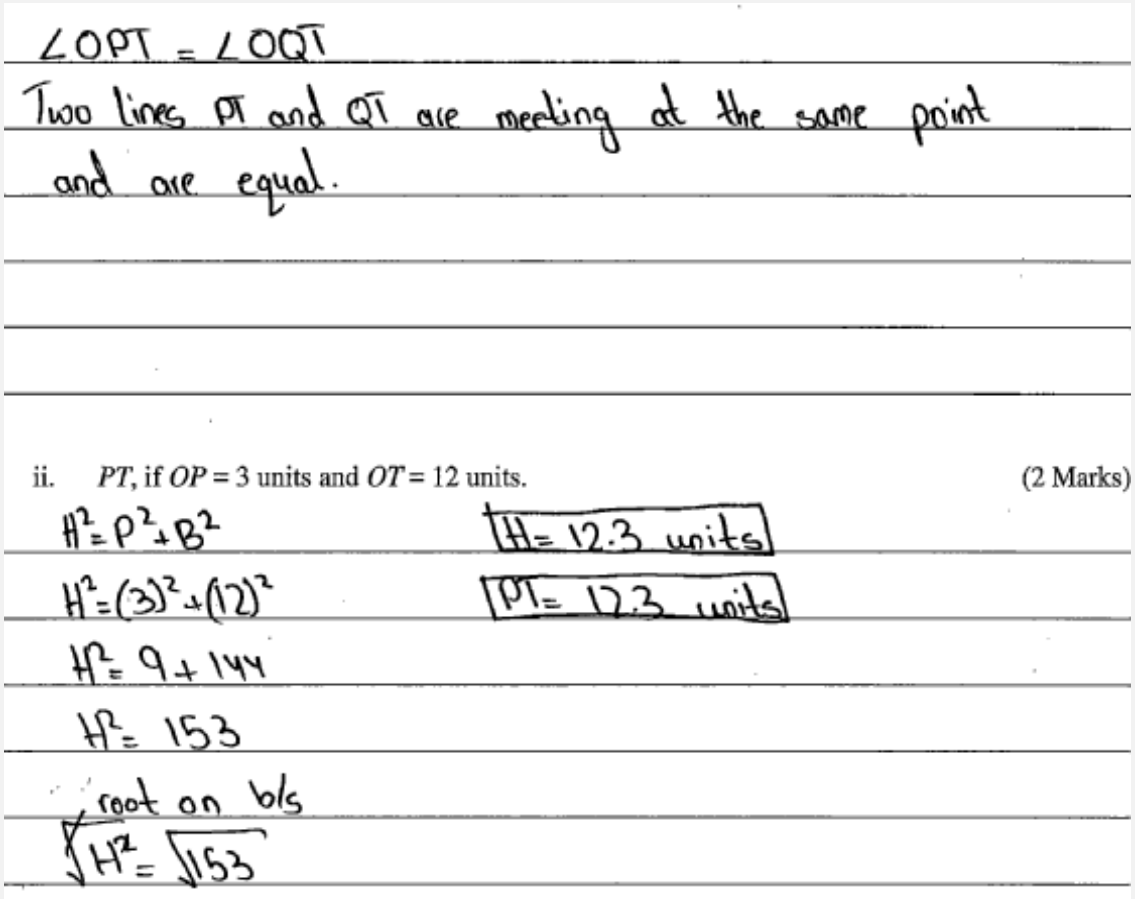
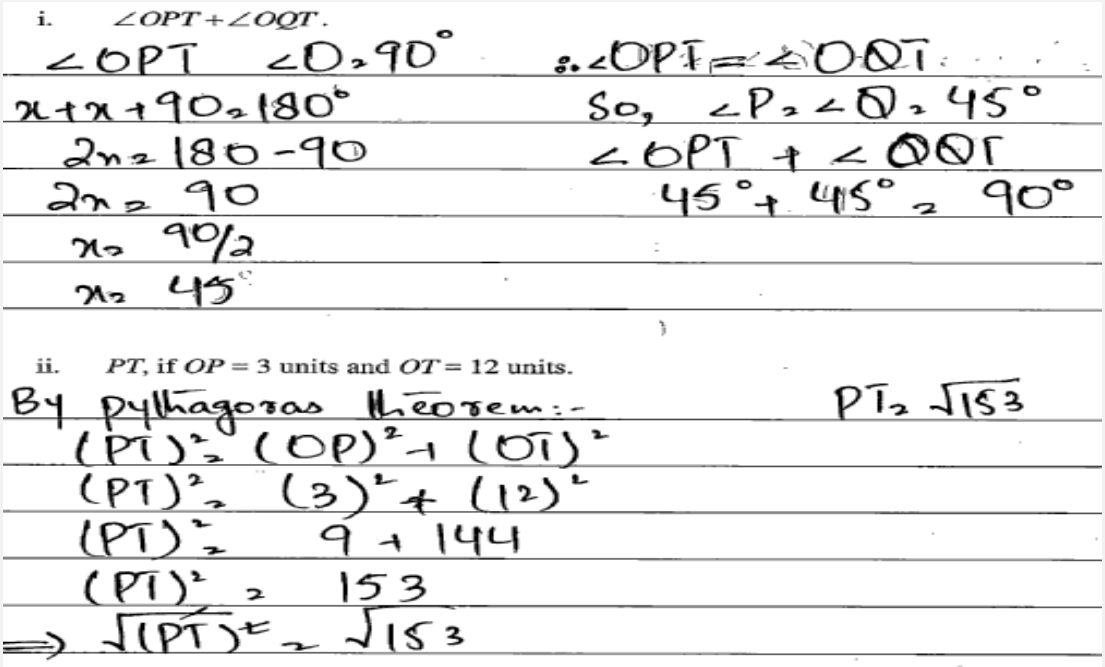
Description of Weaker Responses	Weaker responses showed misconceptions regarding tangents on the circumference of the circle. These candidates were unable to identify the angle between the outer point of the radial segment and the tangent at that point. Due to this reason, some of the candidates were unable to identify the hypotenuse in triangle *OPT*, hence, got incorrect values and lost their marks.
Images of Weaker Responses	**Image (i)**  i. $\angle OPT + \angle OQT$. (2 Marks) $\angle OPT$:- $H^2 = P^2 + B^2$ $H^2 = (12)^2 + (\sqrt{3})^2$ $H^2 = 144 + 9$ $H = \sqrt{153}$ $\angle OQT$:- $H^2 = P^2 + B^2$ $H^2 = (12)^2 + (3)^2$ $H^2 = 144 + 9$ $H = \sqrt{153}$ $\angle OPT + \angle OQT$:- $= 12.3 + 12.3$ $= 24.6$ units **Image (ii)**  $\angle OPT = \angle OQT$ Two lines PT and QT are meeting at the same point and are equal. ii. *PT*, if *OP* = 3 units and *OT* = 12 units. (2 Marks) $H^2 = P^2 + B^2$ $H^2 = (3)^2 + (12)^2$ $H^2 = 9 + 144$ $H^2 = 153$ root on b/s $\sqrt{H^2} = \sqrt{153}$ $H = 12.3$ units $PT = 12.3$ units

Image (iii)



Suggestions for Improvement (Highlighted part)

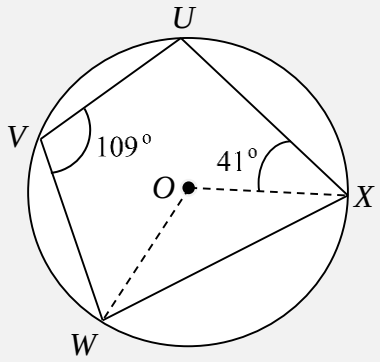
How to Approach SLO	Pedagogy Used for that SLO	Assessment Strategies
• Understand the expectations of the command words • Look at the cognitive level • Identify the content that is required to answer that question (both in terms of understanding of concepts and any skills that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual resources • Think, Pair and Share • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:
While teaching the concept of tangent to a circle, teachers are advised to use the following strategies:
Geometric Exploration: Encourage students to explore tangents and chords on a physical circle. Provide them with circular objects and ask them to draw tangents and chords, measuring angles and lengths to understand the relationships.
Tangent and Chord Applets: Utilise interactive online applets or software that allows students to manipulate circles, tangents, and chords in a dynamic and visual way. This can help them grasp the concepts through hands-on experience.

Hands-On Activities: Organise hands-on activities where students use compasses, rulers, and protractors to draw circles, construct tangents, and measure angles. These physical experiences can enhance understanding.
Real-Life Applications: Connect the concepts of tangents and chords to real-world scenarios, such as car tires, bicycle wheels, or architectural designs. Show how these concepts are used in practical situations.
Geometry Software: Introduce geometry software like GeoGebra or Desmos, which allows students to experiment with geometric constructions involving circles, tangents, and chords.

Question No. 8b

Candidates were given the choice to attempt any ONE out of the two questions: 8a and 8b.

Question Text	In the given diagram, O is the centre of the circle. **NOT TO SCALE**  If $\angle UVW = 109^{o}$ and $\angle UXO = 41^{o}$, then find the value of i. $\angle OXW$. ii. $\angle OXW + \angle OWX$.
SLO No.	26.1.1f
SLO Text	Apply the following theorems to solve related problems (f) the opposite angles of any quadrilateral inscribed in a circle are supplementary.
Max Marks	4
Cognitive Level	A
Checking Hints	i. 1 mark for writing $109^{o} + \angle UXW = 180^{o}$ 1 mark for writing $\angle UXW = 71^{o}$ 1 mark for finding $\angle OXW = 30^{o}$ ii. 1 mark for writing $\angle OXW + \angle OWX = 30^{o} + 30^{o} = 60^{o}$ to get the correct answer
Overall Performance	In question 8b, candidates were required to recall the theorem 'the opposite angles in a cyclic quadrilateral are supplementary'. Applying the theorem would help to find the unknown angles. Most of the candidates attempted this question, out of which fewer of the candidates obtained full marks. However, the common error found was the inability to recall the related theorem to find the unknown angles.

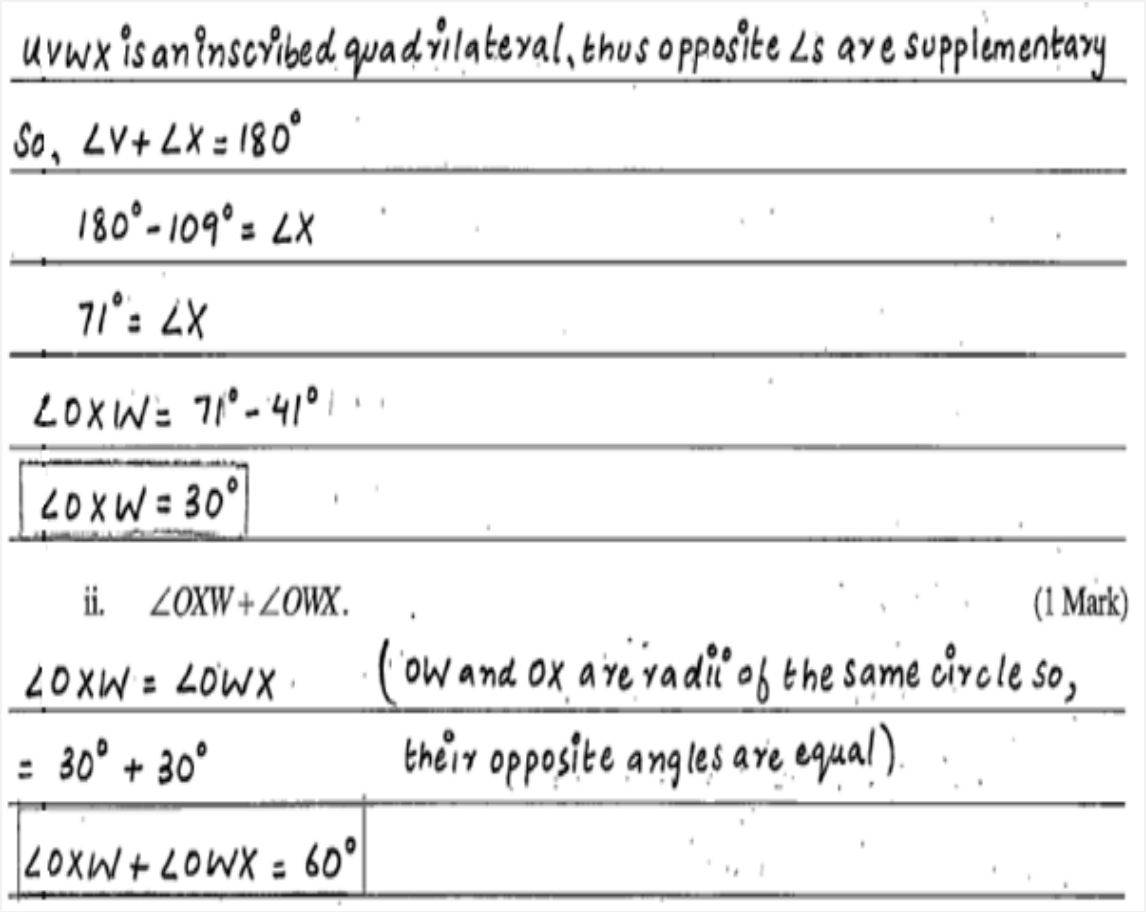
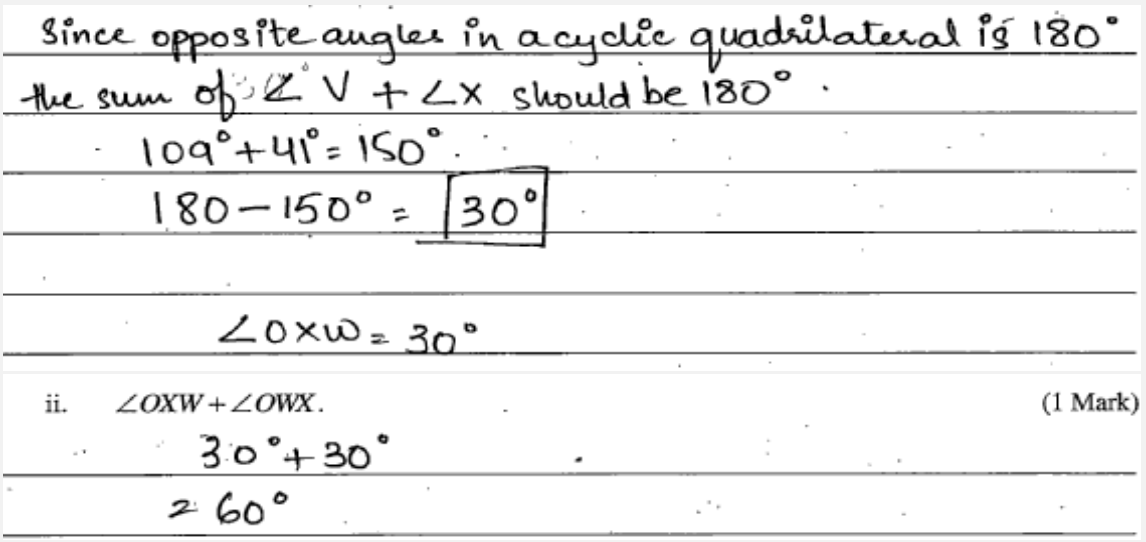
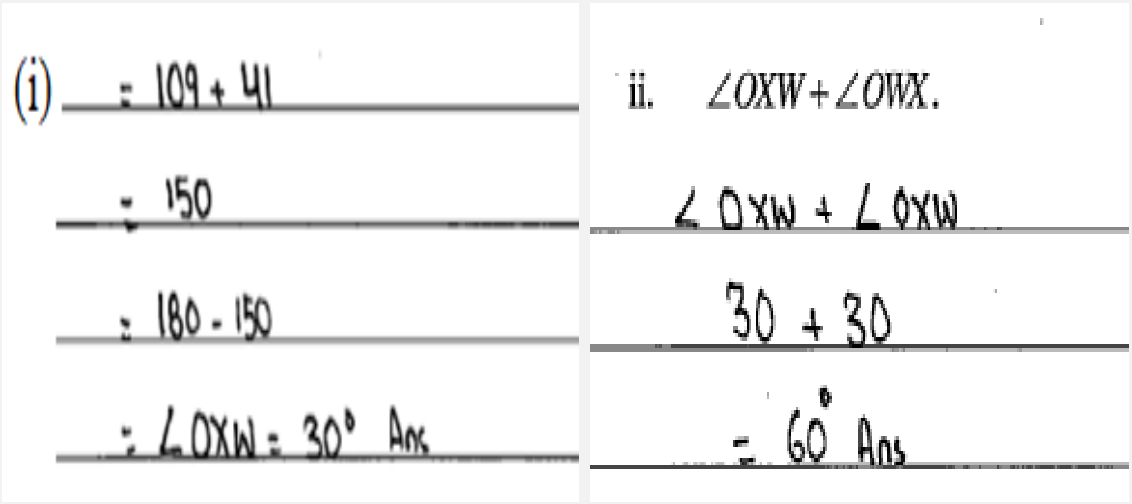
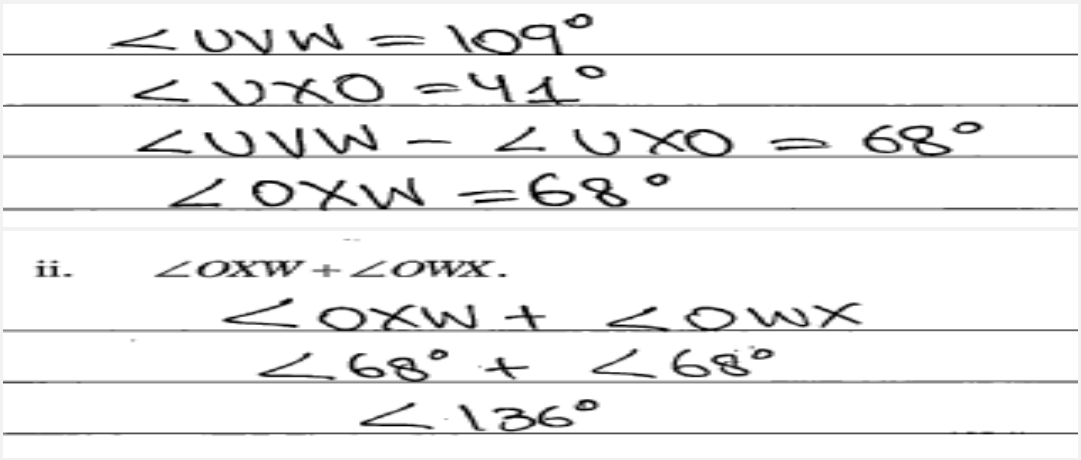
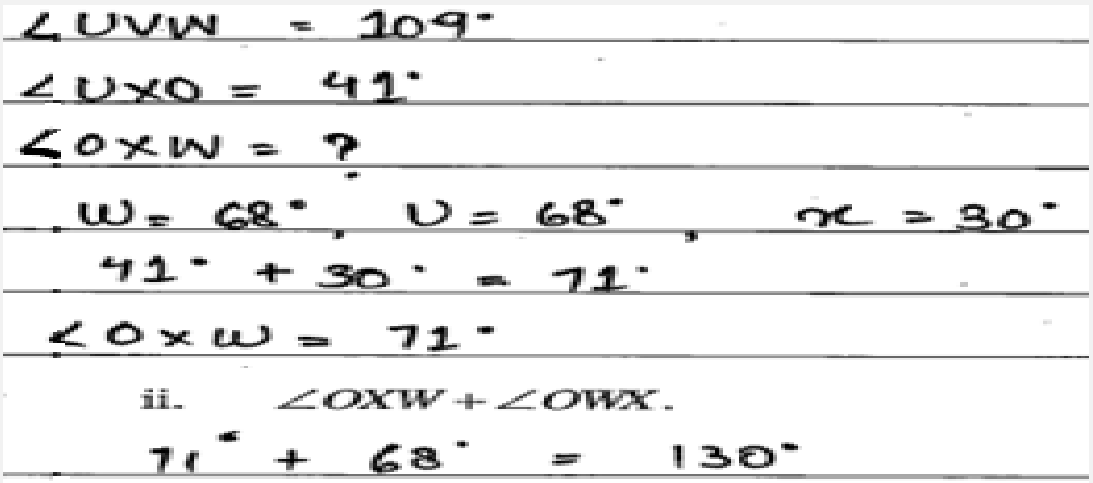
Description of Better Responses	Better responses showed the proper working of the question which indicated that students have clear understanding of the properties of inscribe quadrilateral. In part (i), candidates applied the opposite angle property directly. Some of the candidates calculated angle $\angle UXW = 71^o$ first and then calculated angle $\angle OXW = 30^o$, while others added 109° and 41° first and then subtracted by 180 to get $\angle OXW = 30^o$. In part (ii), candidates added the correct required angles.
Images of Better Responses	**Image (i)**  UVWX is an inscribed quadrilateral, thus opposite ∠s are supplementary So, $\angle V + \angle X = 180^\circ$ $180^\circ - 109^\circ = \angle X$ $71^\circ = \angle X$ $\angle OXW = 71^\circ - 41^\circ$ $\boxed{\angle OXW = 30^\circ}$ ii. $\angle OXW + \angle OWX$. (1 Mark) $\angle OXW = \angle OWX$ (OW and OX are radii of the same circle so, $= 30^\circ + 30^\circ$ their opposite angles are equal). $\boxed{\angle OXW + \angle OWX = 60^\circ}$ **Image (ii)**  Since opposite angles in a cyclic quadrilateral is 180° the sum of ∠V + ∠X should be 180°. $109^\circ + 41^\circ = 150^\circ$. $180 - 150^\circ = \boxed{30^\circ}$ $\angle OXW = 30^\circ$ ii. $\angle OXW + \angle OWX$. (1 Mark) $30^\circ + 30^\circ$ $= 60^\circ$

	Image (iii) 
Description of Weaker Responses	Weaker response showed that the candidates were unable to identify the inscribe quadrilateral in the given diagram. They just found the difference of 41° from 109 ° and supposed it as $\angle OXW$. Some candidates correctly attempted part (i) however, became confused in part (ii) and were unable to observe that triangle $\angle OWX$ as isosceles triangle.
Images of Weaker Responses	**Image (i)**  **Image (ii)** 

Suggestions for Improvement (Highlighted part)

How to Approach SLO	Pedagogy Used for that SLO	Assessment Strategies
• Understand the expectations of the command words • Look at the cognitive level • Identify the content that is required to answer that question (both in terms of understanding of concepts and any skills that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual resources • Think, Pair and Share • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

To improve students' performance in this theorem teachers are suggested to use the following strategies with real life examples.

Real-Life Example - Ferris Wheel: Start by introducing a real-life example like a Ferris wheel. Explain that the cabins on the Ferris wheel can be considered as a quadrilateral inscribed in a circle.

Hands-On Activity - Inscribed Quadrilaterals: Provide students with cut-out circular shapes and several quadrilateral shapes (e.g., rectangles, parallelograms, trapezoids). Let them explore and arrange these quadrilaterals within the circular shape to discover that opposite angles are supplementary.

Interactive Geometry Software: Use geometry software such as GeoGebra to create dynamic visuals. Students can manipulate the quadrilateral within a circle and observe how the opposite angles change while maintaining their supplementary relationship.

Problem-Solving Challenges: Present students with a series of problems that involve inscribed quadrilaterals. Encourage them to apply the concept to find missing angle measures. Use real-world scenarios, like architectural design or navigation, as contexts for these problems.

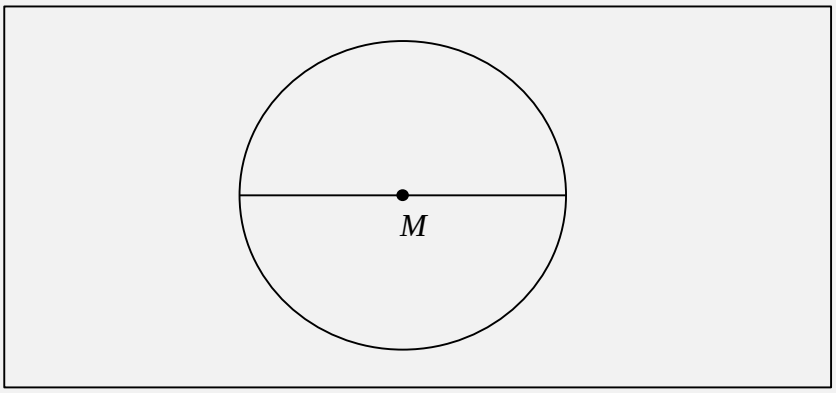
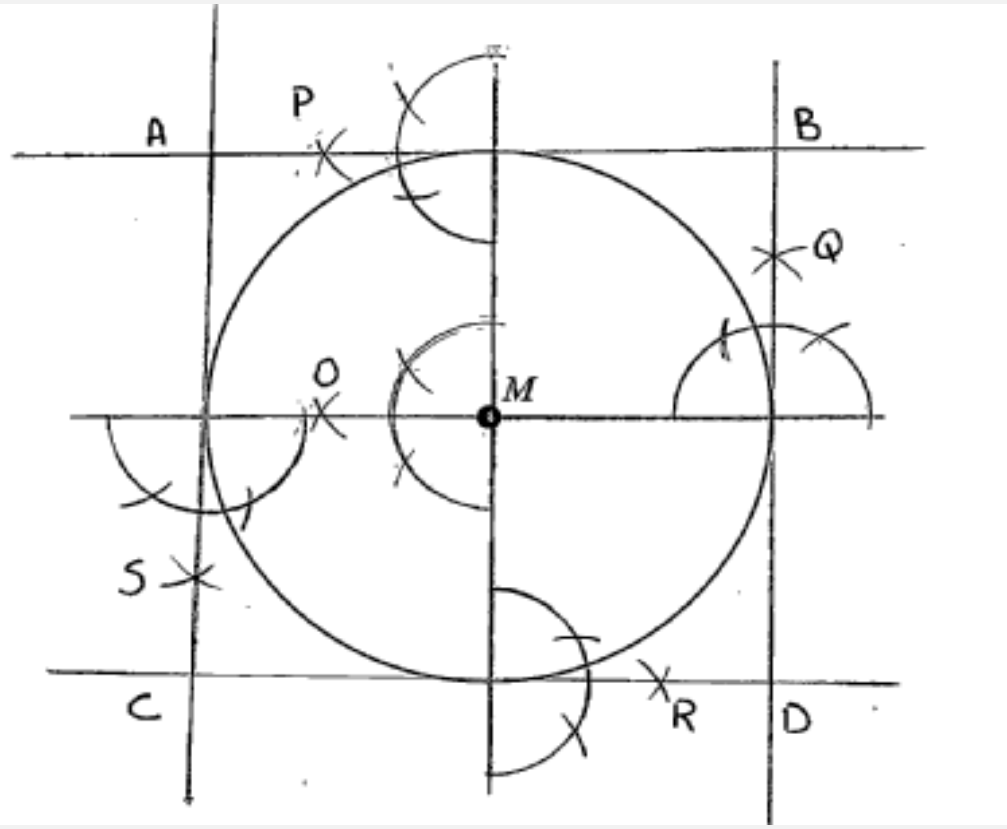
Question No. 9	
Question Text	In the given diagram, draw a circumscribed square around the given circle. 
SLO No.	27.2.6
SLO Text	Draw a circumscribed square about a given circle.
Max Marks	3
Cognitive Level	A
Checking Hints	1 mark drawing the bisector of the given diameter. 1 mark for marking arcs to attain four vertices of the square 1 mark for joining the points to get the required square
Overall Performance	This question was related to construction of circumscribed square for the given circle with the help of compass. This question was well attempted by majority of the candidates.
Description of Better Response	Better responses exhibited the concept of circumscribed square of a given circle and were able to construct the square with proper arcs by using compass.
Images of Better Responses	**Image (i)** 

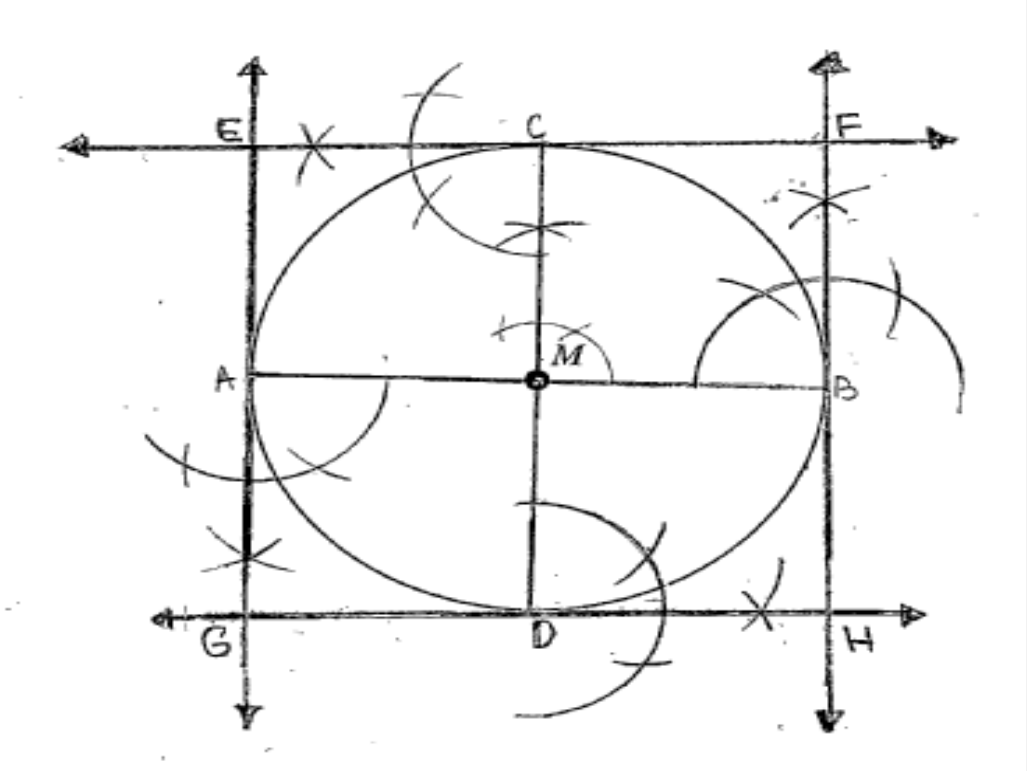
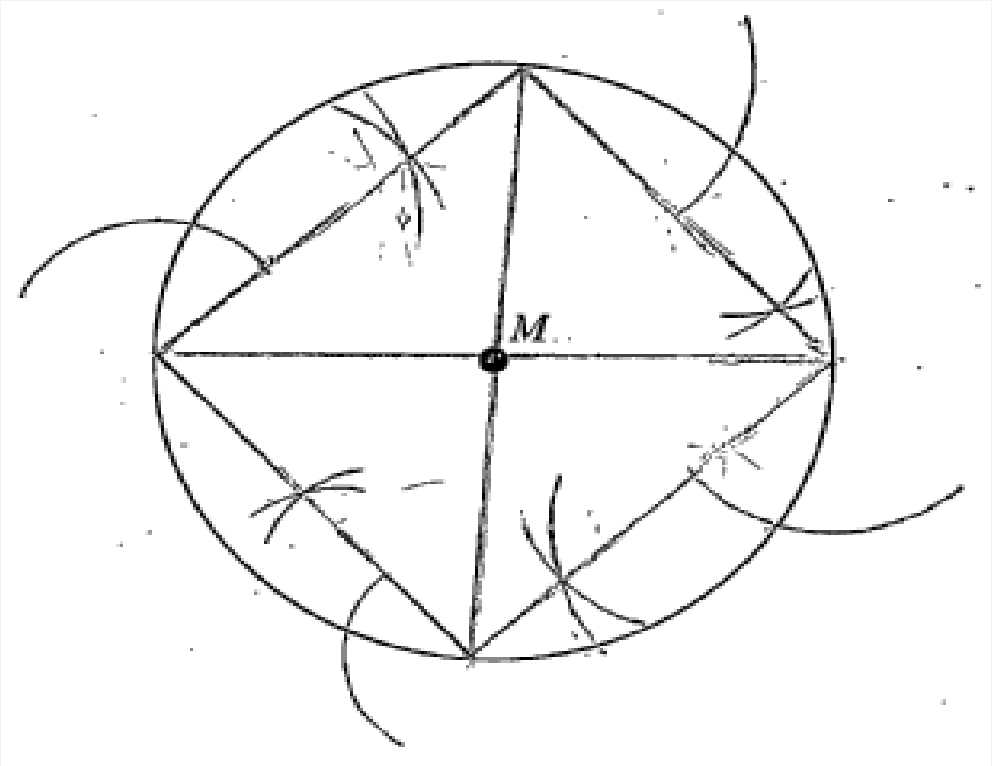
	Image (ii) 
Description of Weaker Responses	Weaker responses showed that some of the candidates did not mark proper arcs to draw circumscribe square. Some of the candidate drew the square without using compass and lost their marks. Some candidates were also unable to differentiate between inscribed and circumscribed square and hence, drew inscribed square.
Images of Weaker Responses	**Image (i)** 

Image (ii)

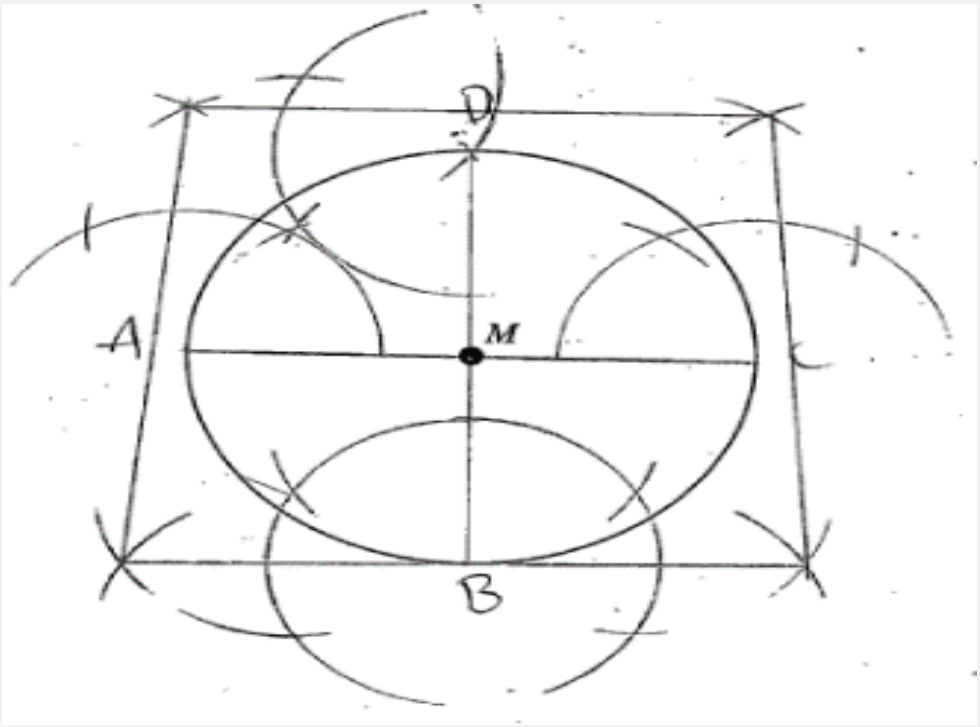
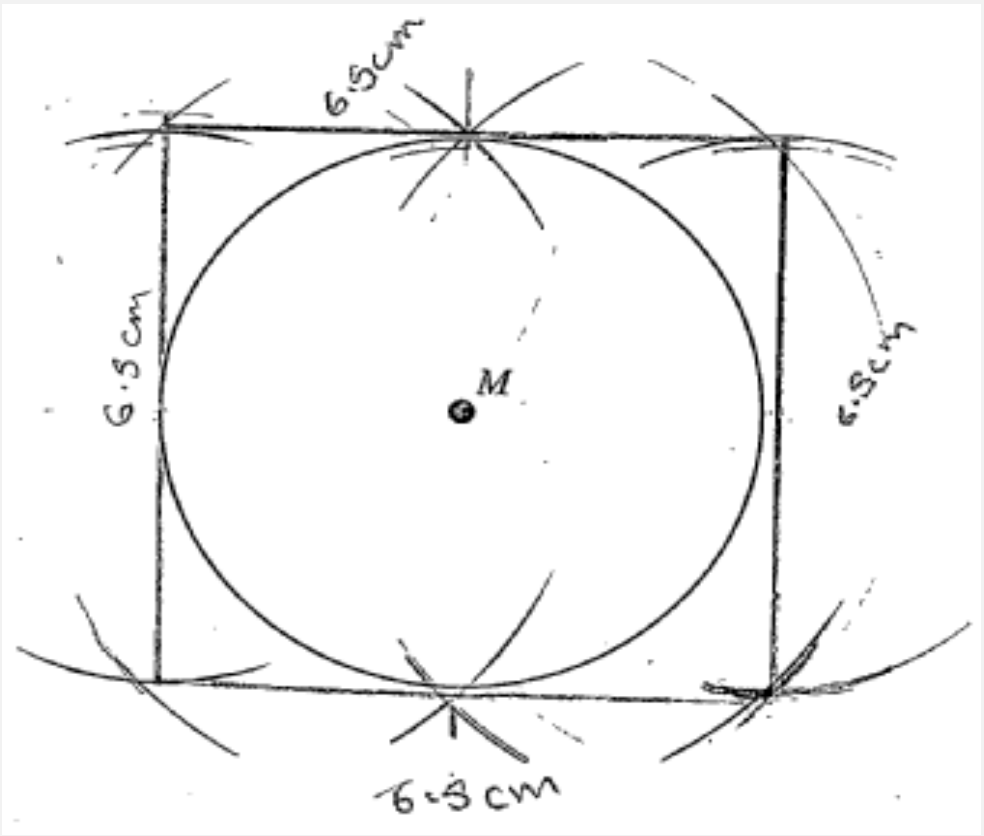


Image (iii)



Suggestions for Improvement (Highlighted part)

How to Approach SLO	Pedagogy Used for that SLO	Assessment Strategies
• Understand the expectations of the command words • Look at the cognitive level • Identify the content that is required to answer that question (both in terms of understanding of concepts and any skills that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual resources • Think, Pair and Share • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

While teaching the concept of circumscribed square, teachers may use the following strategies:

Visual Aids and Demonstrations: Start with a clear, visual demonstration. Use a whiteboard or chalkboard to draw a circle and illustrate the concept of a circumscribed square step by step. You can also use digital tools or software to create dynamic visual demonstrations.

Hands-On Activity: Provide students with compasses, rulers, and paper. Have them practice drawing a circumscribed square around a circle themselves. This hands-on activity reinforces the concept through physical experience.

Interactive Software: Explore interactive geometry software like GeoGebra or Desmos. These tools allow students to manipulate objects on the screen and observe how the square circumscribes the circle as they change its size.

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard
Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.
Example: In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect
Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.
Example: In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone
Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.
Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

Pedagogy: Concept Mapping
Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.
Example: In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources
Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.
Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

Pedagogy: Think, Pair, and Share
Description: A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.
Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange

their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)
Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.
Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration
Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.
Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

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Additionally, we express our gratitude to the esteemed team of reviewers for their constructive feedback on overall performance, better and weaker responses, and validating teaching pedagogies along with suggestions for improvement.
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